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Predicting corporate policies using downside risk: A machine learning approach[☆]

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A B S T R A C T

This paper develops a text-based downside risk measure using corporate annual reports and assesses its ability to forecast future corporate policies. The forward-looking measure dynamically captures adverse firm conditions evolving from economic fundamentals. When the measure is below its sample average, leverage, investment, R&D, employment, and dividends consistently fall. When the measure rises, firms increase cash holdings. The proposed measure also delivers robust and persistent forecasts based on in-sample and out-of-sample LASSO regressions.

1. Introduction

A growing body of work has shown that text-based measures display incremental forecast power in asset pricing and corporate finance. As an example, Hoberg et al. (2014) develop a textual measure of product market fluidity as a proxy for product market risk, and relate it to dividend payouts, share repurchases, and cash holdings.[1] In line with the fast evolving text mining literature, this paper develops a comprehensive measure of downside risk and assesses its incremental predictive power for the dynamics of a variety of corporate policies, including capital structure, investment, R&D, employment, cash holdings, dividend payouts, and stock repurchases. In particular, we use computer linguistics to analyze 76,676 annual reports covering 10,940 firms from January 1994 to September 2015. We then construct a dictionary of 84 keywords that reflect downside possibilities characterizing firms' fundamentals (see the list of our keywords in Table A.1). A risk level measure is defined as the ratio of downside risk-relevant keywords to total word usage in a report, while risk change is defined as the annual change in that ratio.

The newly proposed measure reveals several important merits relative to existing quantitative measures including stock return volatility and earnings volatility. First, the text is extracted from annual report descriptions that must be representative and meaningful to meet regulatory and legal standards, and it integrates managerial risk discussions about corporate and economic fundamentals. Therefore, relative to stock return volatility, the text-based measure is less prone to investor clienteles, behavioral

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[1] Antweiler and Frank (2004) employ Naive Bayes algorithms and find that internet stock messages predict market volatility. Li (2006) develops textual measures of corporate risk and shows that the change in risk exposure predicts future stock return, volatility, and trading volume. Tetlock (2007) and Tetlock et al. (2008) employ Harvard psychosocial dictionary to find that the tones of Wall Street Journal articles and corporate annual reports predict future stock returns and earnings. Loughran and McDonald (2011) construct a dictionary of negative tone words, and relate them to stock returns, trading volume, and unexpected earnings. Loughran and McDonald (2013) develop a text-based uncertainty measure, and examine its implications on post-IPO stock performance. Hoberg and Maksimovic (2015) innovate textual measures of financial constraints.

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biases, short-term market frictions, and investor sentiment.[2] Second, the new measure is forward-looking in capturing perceived business conditions, unlike earnings volatility that reflects historical outcomes. Finally, the text-based measure is constructed from keywords that reflect downside possibilities, while stock return volatility and earnings volatility are derived from second moments capturing both upside and downside prospects.[3,4] The correlation between the text-based risk measure and stock return volatility (earnings volatility) is significant, but only at 7.76% (5.42%), confirming that the text-based measure is potentially informative of downside risk uncaptured by existing proxies.

Compared with existing textual measures, our dictionary of keywords is not limited to the word "risk" per se (Li, 2006), words with negative tones (Loughran and McDonald, 2011), or words characterizing product market competition and threat (Li et al., 2013; Hoberg et al., 2014). Rather, our measure aims to capture a broader set of well-documented risk factors such as competition, credit risk, operational risk, and economic downturn.

The text measure delivers intuitive results: the financial, insurance, and real estate industries display the highest risk exposure during the sample period, while agriculture and utility firms exhibit the lowest risk. At the firm level, less profitable firms with lower credit ratings and fewer tangible assets are associated with higher level of downside risk. The largest risk changes are recorded in the 2008 financial crisis. The textual analysis also helps identify other major changes that firms confront, such as demand shocks, manufacturing overcapacity, industry consolidation (M&As), and increasing volatility in product and raw material prices (see examples of the largest downside risk changes in Table 2). While such results could be expected, we show below that there is a robust and persistent association between the text-based measure and all corporate policies examined.

In the first place, our conventional regression results show that increases in risk are associated with diminishing leverage, capital expenditure, R&D, employment, and dividend payouts and stock repurchases, along with increasing cash holdings in the following year. Strong statistical significance is recorded after controlling for year and industry fixed effects and then implementing robust standard errors clustered at both the firm and year levels in OLS regressions. Furthermore, we find that firms take on defensive actions when downside risk rises and reverse these policies when risk resolves,[5] suggesting that the text measure could capture the dynamics (both the deterioration and resolution) of business conditions. In contrast, the relations between changes in stock return volatility and capital structure, investment, R&D, and employment decisions are mostly insignificant, consistent with the mixed prior evidence related to such corporate decisions.[6]

We show that downside risk changes can predict changes in corporate policies in the longer run. Capital structure, capital expenditure, R&D, employment, and cash holding adjustments could last over three years following risk changes, while adjustments in dividends and stock repurchases last over one to two years. In that perspective, the firm-level evidence reinforces the findings in Bloom (2009) that first-moment shocks exert long-lasting impact on economy-wide investment and employment.

We further show that our new measure has more predictive power than existing text-based and non-text-based proxies based on in-sample and out-of-sample LASSO regressions. Relative to the OLS method, LASSO procedures accommodate a larger set of potential predictors and model structural forms with the added benefit of selecting variables and mitigating overfit. Our results reveal that the text measure is the most frequently-chosen variable by LASSO procedures compared with return and earnings volatilities as well as other textual measures based on existing dictionaries such as Li (2006), Tetlock (2007), Loughran and McDonald (2011), Kravet and Muslu (2013), and Hoberg et al. (2014).[7]

Our evidence is robust to a comprehensive battery of tests. First, we consider alternative downside risk measures based on Item 7 and 7a (Managerial Discussion and Analysis), Item 1 (Business), and Item 1 A (Risk Factor), respectively. The results are qualitatively similar. The results are also robust to alternative weighting schemes for the risk-related keywords, accounting for the potential effects of CEO traits and corporate hedging activities, alternative proxies for corporate policies, excluding financial firms, or excluding the 2007–2008 financial crisis period.

As stated earlier, a large body of work using text analysis is tightly related to ours. We add to this important body of work in the following ways. First, to the best of our knowledge, we are the first to study a comprehensive set of corporate decisions in a unified framework, while the existing literature typically focuses on earnings and stock market performance (e.g., Antweiler and Frank, 2004; Li, 2006; Tetlock, 2007; Kravet and Muslu, 2013; Loughran and McDonald, 2011, 2013), corporate payout policies (Hoberg

[2] Return volatility often reflects factors unrelated to changing fundamentals, such as investor clientele (e.g., investor horizon as in Cella et al. (2013); investor preferences as in Baker and Wurgler (2004)), behavioral trading patterns (e.g., herding, overconfidence, and loss aversion as in Scharfstein and Stein (1990); Barberis Nicholas and Thaler (2003)), short-term market frictions (e.g., market illiquidity as in Amihud and Mendelson, 1986; Pastor and Stambaugh, 2003) and investor sentiment (Baker and Wurgler, 2007; Da Zhi et al., 2015; De Long et al., 1990; Shiller, 1981). Market-based measures are also likely to be the equilibrium of multiple economic forces, thereby casting concerns raised by Qiu and Welch (2006): "How does one test a theory that is about from inputs to outputs with an output measure?"

[3] According to Merriam-Webster dictionary's definition of risk: "Risk is the possibility that something *bad* or *unpleasant* (such as an injury or a loss) will happen". Our focus on the downside feature of corporate risk is, therefore, consistent with the dictionary definition.

[4] Bloom (2009) argues that first and second moment shocks may exert different implications for investment and employment. In particular, second moment shocks may cause management to adopt a "wait-and-see" strategy rather than comprehensive adjustments of corporate policies implied by first moment shocks.

[5] The only exception exists for dividend and stock repurchase policy, where we find that firms tend to reduce their dividends and repurchases when downside risk arises, but they may not immediately increase their dividends and repurchases when such risk resolves.

[6] As an example, Harris and Raviv (1991) conclude in their comprehensive survey that the relation between corporate risk exposure and capital structure is inconclusive. In the same vein, Frank and Goyal (2009) find that stock return volatility constitutes only a "minor factor" in explaining book leverage, and Panousi and Papanikolaou (2012) show that the risk-investment relation is economically weak.

[7] Note that we also incorporate textual variables based on positive tone use (Tetlock, 2007; Loughran and McDonald, 2011). The results indicate that the positive tone use seems to have some predictive power in predicting capital structure, cash holdings and dividend policies. But the prediction performance of these measures is generally weaker than the measures based on negative tone use and product market threat. Overall, the down-side risk measure is the most consistent and robust predictor among all textual measures examined.

et al., 2014; Bodnaruk et al., 2015), or investment decisions (Hoberg and Maksimovic, 2015). Second, our measure captures a broader set of risk factors arising from industry competition, macro conditions, financial market, counterparties, and day-to-day operations. The results show that although measures based solely on negative words (Loughran and McDonald, 2011) or product market threat (Hoberg et al., 2014) have some predictive power in certain corporate decisions, our new measure is the most consistent and robust predictor in terms of all corporate polices examined.

Our paper complements the fast-growing literature on machine learning applied to predicting future stock returns. Gu et al. (2020) apply a comparative analysis of machine learning methods to measuring asset risk premia. Avramov et al. (2019) assess the economic significance of machine learning routines, while Avramov et al. (2020) implement LASSO and Ridge regressions to predict future returns by deviations of accounting items from their means. In our application, we use LASSO procedures to examine the in-sample and out-of-sample predictability of corporate policies. Our LASSO estimations overall indicate robust in-sample and out-of-sample prediction performance, especially for capital structure and payout policies.

Our paper also adds to the literature on risk and corporate polices. By constructing a forward-looking downside risk measure based on firm fundamentals, we obtain robust and persistent empirical evidence consistent with economic theories and comprehensive surveys in the field.[8] Our new measure particularly outperforms the market-based return volatility in predicting longer-term leverage and investment related decisions, where existing empirical evidence is quite weak (e.g., Frank and Goyal, 2009). Finally, our findings are consistent with the notion that corporation decisions are associated with downside risk evolving from firm fundamentals, rather than short-term market factors such as investor sentiment, short-term market frictions, and behavioral trading patterns.

The remainder of the paper is organized as following. Section 2 presents theoretical background to motivate our text-based downside risk measure. Section 3 introduces the construction and characteristics of the risk measure. Sections 4 and 5 present our data and empirical findings, respectively. Section 6 implements a battery of robustness checks. Section 7 concludes the paper.

2. Theoretical background

In this section, we provide a brief discussion of the related literature, and motivate the notion that risk measure could potentially predict certain corporate policies.

A large body of corporate finance theory (e.g., Leland, 1994; Hennessy et al., 2007; Bolton et al., 2011, 2013) and field surveys (e.g., Brav et al., 2005; Lintner, 1956) suggest that firms adopt more conservative investment and financial policies in response to increasing risk exposure. We thus follow this strand of literature to examine four major types of corporate policies.

First, the trade-off theory (Modigliani and Miller, 1958; Merton, 1974) suggests that increases in risk are likely to exert a significant effect on leverage. The theory asserts that firms choose capital structure to balance the benefits of debt financing and the direct and indirect costs associated with financial distress. Rising risk essentially leads to higher default probability and thus to increasing borrowing costs, which could induce corporations to reduce debt. In addition, more volatile cash flows reduce the likelihood that tax shields could be fully utilized.[9] Based on this theoretical framework, we study the predictability of firm capital structure using the new down-side risk measure.

Second, as noted earlier, Bloom (2009) models the impact of economic shocks and predicts that first moment shocks lead to long-lasting reductions in investment and employment. In addition, at the micro-level, Bolton et al. (2011) and Bolton et al. (2013) show theoretically that corporate investment should decline when uncertainty rises.[10] In this paper, we extend these existing studies to investigate a variety of corporate investment decisions ranging from capital expenditure and employment to R&D and advertising expenditure.

Third, we examine corporate cash holding decisions as past work shows that firms primarily hold cash for precautionary motives to hedge against adverse risk shocks. Rising risk, therefore, should be associated with a reduction in cash holdings. For instance, Bates et al. (2009) attribute the rising cash holdings of US manufacturing firms to increasing cash flow risk. Hoberg et al. (2014) and Gao et al. (2014), among other studies, find that cash holdings of both public and private firms in the U.S. are positively related to their risk exposure.

Fourth, classic field surveys (Lintner, 1956; Brav et al., 2005) indicate that risk is an important consideration for managers in making payout decisions. Empirical evidence generally supports this view. For example, Hoberg and Prabhala (2009) and Hoberg et al. (2014) show that firms with higher risk exposure are associated with less dividend payouts and stock repurchases. Bliss et al. (2015) also document significant reductions in dividends and repurchases during the global financial crisis.[11] Following this literature, we use changes in risk to predict corporate payout policies.

8 See Section 2- for a summary of the related literature.

9 The existing empirical evidence on risk and capital structure, however, is mixed (Harris and Raviv, 1991). Many past studies focusing on earnings volatility (e.g., Titman and Wessels, 1988; Kester, 1986) or return volatility (e.g., Welch, 2004; Frank and Goyal, 2009) document a weak relation between risk and leverage ratio.

10 The overall theoretical conclusions regarding the relation between uncertainty and corporate investment, however, are ambiguous as the sign may depend on the shape of adjustment costs and other factors. Empirical findings on the sign and significance of the investment-stock return volatility relation are also sensitive to the estimation method. Please see Panousi and Papanikolaou (2012) for a comprehensive review of the theoretical and empirical literature.

11 Floyd et al. (2015), however, find that banks have a higher propensity to pay dividends and resist cutting dividends as the 2007–2008 financial crisis begins, implying that banks use dividends to signal financial strength.

3. Measuring downside risk

3.1. Textual analysis

We develop a web crawling program to collect 10-K corporate filings from the Security and Exchange Commission's EDGAR website. Our sample spans the period from January 1994 to September 2015. Merging the corporate filings with CRSP and COMPUSTAT databases leaves us with 76,676 filings corresponding to 10,940 distinct firms. To analyze the textual content of the filings, we delete numbers, tables, graphs, propositions, articles, and pronouns.[12] We further decompose the texts into word stems, hence, our analysis is based on the underlying meaning of words regardless of their different tenses or formats. We build upon existing textual analysis literature (Li, 2006; Loughran and McDonald, 2011, 2013; Li et al., 2013) to construct a dictionary of keywords characterizing managerial perception of downside risk regarding firm fundamentals. The dictionary is composed of 84 downside risk-related keywords (see Table A.1 for the key word stems and keywords and A.2 for examples of sentences from the annual reports containing each key word stem).

Our keyword list starts with common words with negative tones to highlight the downside feature of corporate risk. In particular, we borrow from the negative word list in Loughran and McDonald (2011, 2013) and select among the list those words that appear most frequently in our sample.[13] Next, our keywords include various formats of the word "risk" per se (e.g., "risky", "risks") as in Li (2006). To make the measure comprehensive and informative,[14] we further include keywords describing specific types of firm risk, such as "competition", "downgrade", "downturn", "crisis", all of which are well documented risk characteristics in the finance literature.[15]

As noted in the introduction, we define risk level as the ratio of downside risk-relevant keywords to total words in those reports (excluding propositions, articles, and pronouns). Risk change is the annual change in the risk level. We also consider two dummy versions of risk change. A dummy for rising risk is equal to one when firm experiences rising risk in a magnitude larger than the median increase in risk and zero otherwise. A dummy for diminishing risk is equal to one when firm experiences decreasing risk in a magnitude larger than the median decrease in risk and zero otherwise.

3.2. Characteristics of the text-based downside risk measure

Panel A of Table 1 presents the distribution of average risk levels by industry groups. We divide the sample into ten broad industry groups based on the SIC codes and compute the average downside risk for all groups. Risk levels are typically different across industries. Finance, insurance, and real estate groups exhibit the highest risk, consistent with the notion that the financial sector plays a crucial role in consolidating and managing risks emerging from real activities. In contrast, firms in the utility and agriculture industry are associated with lower level of risks due to stringent regulations and various subsidy and insurance programs.

Panel B of Table 1 presents the distribution of the text-based risk measure by firm characteristics including credit ratings, profitability (EBIT/Asset), tangibility, and existing risk measures. We find that less profitable firms face higher downside risk levels, consistent with the notion that such firms are more likely to be financially constrained and thus are prone to liquidity risk related to external financing. Similarly, firms with low credit ratings and less tangible assets have higher financing costs in the external capital markets and thus face higher level of downside risk.

Panel C presents the correlation between the new measure and macroeconomic variables. Our list of keywords not only characterizes firm-specific risks (e.g., "competition") but also represents how firms view macroeconomic conditions (e.g., downturn, crisis). The results in Panel C reveal that on the one hand, the aggregate text-based downside risk measure is indeed negatively correlated with the Treasury yield, S&P 500 index return, and the industrial production growth, and positively correlated with VIX and the default spread. On the other hand, the correlation coefficients are relatively low, suggesting that the new measure can potentially capture information about the economy incremental to existing predictive macroeconomic items.[16]

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12 Propositions, articles and pronouns are considered non-meaningful words in standard textual analysis.

13 For example, our dictionary includes words such as " loss" and "losses", which are also the most frequently occurring negative words in the sample of corporate reports in Loughran and McDonald (2011).

14 The word "risk" itself is subject to the criticism of being boilerplate (Kravet and Muslu, 2013).

15 We start with listing word stems that appear most frequently (i.e., more than ten times) in the full sample of 10-K filings, and then select from them the keywords describing specific types of firm risk in the following way: First, each researcher is assigned to read 100 randomly selected 10-K reports representing different industries and years. Every researcher judges independently whether a specific word stem reflects a type of firm risk in the context. A word is included in the preliminary dictionary if and only if all three researchers agree so based on their independent judgment. Words in the preliminary dictionary are then checked for consistency with their definitions in the Merriam-Webster dictionary. Further, we solicit opinions from other financial researchers and professionals, and eliminate words that might be controversial. For example, words such as "mergers", "acquisitions" and "fluctuate" are dropped from our dictionary as these words might not necessarily convey downside possibilities. In general, our methodology tilts toward the conservative side in developing the dictionaries of risk-relevant words.

16 Upon incorporating these macro variables into our LASSO predictions of corporate policies in Section 4, we find that S&P 500 index return and industrial production growth have certain predictive power, but our text-based measure is a substantially stronger predictor compared with these macro variables.

Table 1
Characteristics of the text-based downside risk measure.

Panel A: Distribution of downside risk by industry groups

Industry group	Obs.	Mean	Range	
			Min	Max
Agriculture, Forestry, and Fishing	154	0.84%	0.07%	1.57%
Mining	2,928	0.94%	0.10%	2.20%
Construction	860	1.01%	0.04%	1.99%
Manufacturing	29,697	1.00%	0.02%	2.89%
Transportation, Communications, Electric, Gas, and Sanitary Services	6,208	0.87%	0.01%	2.49%
Wholesale Trade	2,404	0.90%	0.04%	2.26%
Retail Trade	4,434	0.87%	0.04%	2.10%
Finance, Insurance and Real Estate	17,579	1.32%	0.03%	3.89%
Services	11,922	1.06%	0.03%	2.62%
Conglomerates	490	1.06%	0.23%	2.35%
Total	76,676	1.06%	0.01%	3.89%

Panel B: Distribution of downside risk by firm characteristics

Firm characteristics	Mean Risk Level		
	Above-the-median Firms	Below-the-median Firms	Difference (1)–(2)
	(1)	(2)	(3)
Return Volatility	1.083%	1.033%	0.050% ***(14.54)
Cash Flow Volatility	1.112%	1.104%	0.008% ***(2.00)
Credit Ratings	0.985%	1.063%	−0.078% ***(−12.07)
EBIT/Asset	0.897%	1.220%	−0.324%*** (−104.55)
Tangibility	0.903%	1.211%	−0.308% ***(−99.94)

Panel C: Correlation with Macro variables

		1	2	3	4	5	6
1	Risk Level	1.000					
2	S&P 500 Return	−0.131***	1.000				
3	Industrial Production Growth	−0.101***	0.594***	1.000			
4	VIX	0.034***	−0.555***	−0.618***	1.000		
5	Default Spread	0.228***	−0.697***	−0.777***	0.624***	1.000	
6	Treasury Yield	−0.450***	0.259***	0.298***	−0.172***	−0.489***	1.000

This table describes the text-based downside risk measure. Panel A presents average, min, and max measures for ten industry groups. Panel B considers above- and below-median size, market-to-book, credit rating, earnings, stock return volatility and cash flow volatility firms. Reported are the average, difference in average, and its *t*-statistics (in parentheses). Panel C presents pairwise correlation coefficients with key macro variables for all firm-year observations. Coefficients marked with *, **, and *** are significant at the 0.1, 0.05, and 0.01 levels, respectively. The sample consists of U.S. listed companies that filed 10-K reports over the period of 1994 January and 2015 September. Risk level is the ratio of downside risk-related keywords to total meaningful words in 10-K files (excluding propositions, articles, and pronouns). Definitions of all variables are provided in Appendix C.

3.3. Examples of large downside risk changes

To better comprehend the proposed risk measure, Table 2 lists examples of firms with the largest downside risk change.[17] Our results highlight the severe impact of the 2007–2008 global financial crisis, as 11 out of the 15 extreme risk changes are attributable to that period. In particular, seven commercial banks and insurance companies (e.g., First Financial and BB&T) faced substantial increases in credit risk and liquidity risk. Construction, TV and radio broadcast, and hotel industries were also challenged by severe liquidity concerns, in addition to sharp drop in their product demand (e.g., the broadcast companies were impacted by diminishing advertising expenses, especially by the financial sector). Other sources of extreme risk changes include large demand shocks from downstream industry, industry consolidation, manufacturing overcapacity, and increasing volatility in product and raw material prices.

While extreme changes seem to concentrate in the financial industry and during the crisis period, we show, in Section 5, that our main results are not driven by financial firms or by the global financial crisis. In particular, similar results are obtained as we exclude financial firms from our sample, when we restrict our sample to manufacturing firms, or when we exclude the years of 2007 and 2008.

4. Data and summary statistics

Our sample consists of 76,676 observations of risk changes, involving 10,940 distinct firms spanning 68 two-digit SIC industries. Policy variables, i.e., leverage, capital expenditure, employment, R&D, cash holdings, dividend payouts, and stock repurchases, are

[17] We acknowledge that our risk change measure is a proxy developed based on changes in risk-related word counts in 10Ks rather than a proxy for real exogenous shocks.

Table 2
Examples of large downside risk changes.

Rank	Fiscal year	Company name	Industry group	Risk change factors
1	2005	SUPERIOR INDUSTRIES INTL	Industry Manufacturing	Reductions in the forecast of new vehicle productions by customers; continued consolidation of the automotive industry; global pricing pressure driven by competitors and cost-cutting initiatives of customers; the need to seek a buyer for the aluminum suspension components business due to significant losses incurred by this invest project; whether the cost-cutting initiatives could be achieved.
2	2009	FNB UNITED CORP	Financial Services	An increase in nonperforming real estate loans; incapability of renewing or accepting brokered deposits without prior regulatory approval and the possibility of paying higher insurance premiums to the FDIC due to decline in the bank's capital position; inability to access the capital markets.
3	2008	FIRST FINL BANCORP INC/OH	Financial Services	Massive write-offs due to credit performance of real estate related loans; inability to access capital because of the tightening of credit market; impairment of goodwill due to unprecedented market volatilities and disruptions; credit risk imposed by the default events of financial institutions.
4	2007	D R HORTON INC	Construction	Declines in demand for new homes; elevated sales cancellation rate, reduction in availability of mortgage financing; declines in profit margin because the company offers higher levels of incentives and price concessions in attempts to stimulate demand.
5	2009	CITIZENS COMMUNITY BANCORP	Financial Services	Deteriorating credit quality; ability to maintain the required capital levels and adequate source of funding and liquidity; further write-downs in residential mortgage-backed securities portfolio; ability to implement the cost-savings initiatives; potential impairment of investment securities, goodwill, and other intangible assets; high volatility of stock price since the stocks are thinly traded.
6	2009	BB&T CORP	Financial Services	Credit deterioration related to the commercial real estate and construction loan portfolios of the newly acquired Colonial Bank and the residential mortgage loans of the bank itself; the ability to expand into the new areas after the acquisition of Colonial Bank; decreases in real estate values, primarily in Georgia, Florida and metro Washington, D.C..
7	2009	SOUTH FINANCIAL GROUP INC	Financial Services	A series of risk factors related to a restatement process: downgrades in credit ratings; acceleration of public debt securities and other debt arrangements due to inability to comply with certain reporting covenants; material weaknesses in internal control over financial reporting; not able to access the public capital markets until all its filings with the SEC are up to date; incapability of attracting and retaining key employees.
8	2004	DORAL FINANCIAL CORP	Financial Services	Difficulty in obtaining additional borrowings or issuing additional equity due to market conditions and recent downgrades of credit ratings; subject to regulatory enforcement actions if not adequately capitalized; failure to comply with the Nasdaq one-dollar minimum bid price requirement; a net increase in Federal income tax if there is an "ownership change" due to new equity issuance or other events; credit losses and impairment charges.
9	2008	FIFTH THIRD BANCORP	Financial Services	Systematic risk faced by the entire industry; increases in competition due to recent consolidation of the financial industry; inability to hire or retain the most qualified senior managers due to the CPP's restrictions on the compensations of senior managers.
10	2002	PROGRESS ENERGY INC	Utility	Energy crisis in California during 2001; the recent volatility of natural gas prices in North America; increased amount of public and regulatory scrutiny due to the bankruptcy filing by the Enron Corporation and recently discovered accounting irregularities of certain public companies; downgrades of senior unsecure debt by S&P and Moody's; drought conditions and related water restrictions in the southeast United States.
11	2002	FIBERCORE INC	Materials and Related Products	Manufacturing overcapacity due to decreased growth in telecommunication industry; a significant and continuing downturn in the South American market; intense competition with several competitors having significantly greater resources and associated pricing pressure; the ability to achieve the cost reduction plans; low cash reserves and limited ability to gain additional capital; possibility of being delisted from Nasdaq due to failure to comply with the one-dollar minimum bid price requirement.
12	2007	AMBAC FINANCIAL GROUP INC	Financial Services	Inability to write new financial guarantee business due to a downgrade of financial strength rating by S&P; an increase in borrowing costs due to downgrades of long-term credit rating by multiple rating agencies; a substantial increase in credit risk related to residential mortgage backed securities and CDOs of ABS; potential disruptions caused by decisions to suspend and discontinue certain business.

(*continued on next page*)

Table 2 (*continued*).

Rank	Fiscal year	Company name	Industry group	Risk change factors
13	2009	SALEM COMMU-NICATIONS CORP	Tele-communications	A significant decrease in advertising by customers in financial services and automotive industries; impairment of broadcast licenses, mastheads and goodwill balances due to increased cost of capital and a decline in the estimated terminal or exit values as a result of industry wide declines in radio station transaction multiples and magazines; ability to integrate the operations and management of two newly acquired radio stations; high credit risk due to substantial previous and new debt obligations.
14	2008	JOURNAL COM-MUNICATIONS	Tele-communications	Impairment of goodwill, tv and radio broadcast licenses, other intangible assets and property, plant and equipment due to deteriorating market conditions and a further decline in the stock price; the adverse impact of changing economic and financial market conditions on liquidity and the availability of capital; the possibility of violating the financial covenants of revolving credit facility; decreases in advertising spending in automotive industry, political advertising and professional sports contracts.
15	2009	MARRIOTT INTL	Services	Significantly reduced demand for hotel rooms and timeshare products around the world; the growing risks of doing business internationally due to growing significance of operations abroad; a downgrade of long-term debt ratings by the three major agencies in 2009; the anticipation that many of the jurisdictions in which the company does business will review tax and other revenue raising laws in response to recent economic crisis; weakened sales of timeshare loans due to disruptions in the credit markets; significant restructuring costs and impairment charges of the timeshare segment.

This table describe 15 examples of the largest downside risk changes. The sample consists of U.S. listed companies that filed 10-K reports over the time of 1994 January and 2015 September. Firms are sorted in descending order based on downside risk change.

from COMPUSTAT. Firm-level control variables, all of which are based on past related work, such as stock returns, credit ratings, and sales growth, are from CRSP and COMPUSTAT. The macro control variables including option-implied volatility (*VIX*), industrial production growth, and one-year maturity Treasury bill yields are from the website of the Federal Reserve Bank of St. Louis. We use various word dictionaries developed by previous studies (Li, 2006; Kravet and Muslu, 2013; Loughran and McDonald, 2011, 2013; Tetlock, 2007; Hoberg et al., 2014) to construct alternative proxies for downside risk.

We describe the construction of data, including textual variables and corporate policy and control variables at the firm and economy-wide levels, in Appendix B. Detailed variable definitions are presented in Appendix C. The descriptive statistics of key variables are reported in Table 3. All continuous variables, except for macro variables, are winsorized at the 1st and 99th percentiles level. The median and standard deviation of book leverage are 55.03% and 27.45%, respectively. The median active change in book leverage is 0.12%. The median annual percentage changes in capital expenditure, employment, R&D, and advertising expenses are 1.77%, 1.35%, 2.41%, and 3.95%, respectively.

The median change in the ratio of cash to assets is −0.03%, with a standard deviation of 8.09%. About 1%–2% of the firms in the sample initiate or omit dividends, while 28% of the firms increase dividends and 9% of the firms shrink dividends. Further, 32% of the firms are engaged in stock repurchase whose value exceeds 1% of total assets, consistent with the recognition that stock repurchases have become a more popular form of payouts (e.g., Bliss et al., 2015; Floyd et al., 2015). Overall, statistics of our key variables are similar to those reported in past work.

5. The Empirical Evidence: Predicting Changes in Corporate Policies

This section examines whether the text measure is useful in predicting changes in leverage, investment, employment, R&D, cash holdings, and payout policies. To start with, we apply the OLS (logistic) panel regression analysis to examine corporate policies on leverage, investment, employment, R&D, and cash holdings (dividend payouts and repurchases), respectively.

The policy regression is formulated as

$$\triangle \mathrm{POLICY}_{i,t+1} = \alpha + \beta_1 \mathrm{RISKCHANGE}_{i,t} + \beta_j \mathrm{CONTROL}_{i,j,t} + \varepsilon_{i,t}, \tag{1}$$

where i is a firm-specific subscript, $\triangle \mathrm{POLICY}_{i,t+1}$ represents future annual change in various corporate policies, and $\mathrm{RISKCHANGE}_{i,t}$ denotes risk change. Control variables for each distinct policy are listed in the data section. Since managers consider different factors in making different corporate decisions, we follow existing work to account for a unique set of control variables in examining each type of corporate policy. As such, the number of observations varies across corporate decisions due to a unique collection of controls employed. We further control for the year and industry fixed effects in all regressions. Since large risk changes can reflect changes in aggregate fundamentals, standard errors may be correlated across firms. We therefore compute standard errors clustered by both firm and year throughout the paper to address that concern (Peterson, 2009).

Next, we apply LASSO regressions to assess the in-sample and out-of-sample incremental predictive power of the text measure in the presence of an exhaustive list of predictors, including those based on dictionaries developed by previous studies. As stated earlier, the LASSO approach has several merits. First, it does not impose a structure form on the model, and the high-dimensional

Table 3
Summary statistics.

Variable	Obs.	Mean	Median	Std. Dev.	Skewness	Kurtosis
Panel A: Text-based Variables						
Total Words	76,676	21,410	19,468	12,742	0.86	3.63
Downside Risk Words	76,676	251	208	207	1.48	5.85
Risk Level	76,676	1.06%	1.04%	0.45%	0.50	3.25
Risk Change	58,607	0.04%	0.02%	0.16%	0.93	10.67
Rising Risk	58,607	0.28	0.00	0.45	0.96	1.91
Resolving Risk	58,607	0.22	0.00	0.41	1.38	2.91
Panel B: Corporate Policy Variables						
Book Leverage	76,676	55.15%	55.03%	27.45%	0.14	2.37
Active Change in Book Leverage (dlev)	57,442	0.09%	0.12%	8.61%	−0.88	8.97
% Change in Debt (%dDebt)	58,585	19.84%	5.72%	60.93%	3.97	22.68
% Change in Equity (%dEquity)	57,441	14.03%	1.23%	54.49%	4.58	28.04
Capital Expenditure (millions)	71,713	113.63	8.91	354.68	5.08	30.96
% Change in Capital Expenditure (%dcapx)	51,439	28.18%	1.77%	134.64%	3.86	21.77
Employment (thousands)	74,085	6.90	0.98	17.99	4.61	26.54
% Change in Employment (%demp)	55,360	7.77%	1.35%	26.93%	2.46	13.22
R&D Expense (millions)	76,676	27.24	0.00	110.43	6.28	45.08
% Change in R&D Expense (%dxrd)	22,694	5.02%	2.41%	29.91%	2.73	15.30
Advertising Expense (millions)	76,676	15.21	0.00	69.84	6.35	45.45
% Change in Advertising Expense (%dxad)	19,540	11.08%	3.95%	65.44%	3.06	18.29
Cash (millions)	76,665	289.24	35.79	949.21	5.75	38.90
Cash/Assets	76,665	18.00%	7.96%	22.48%	1.67	5.03
Change in Cash/Assets (dcash)	58,596	−0.39%	−0.03%	8.09%	−0.41	7.14
Dividends (per share)	76,357	0.33	0.00	0.60	2.39	8.94
Dividend Initiation	58,357	0.02	0.00	0.15	6.39	41.77
Dividend Omission	58,357	0.01	0.00	0.12	8.44	72.26
Dividend Increase	58,357	0.28	0.00	0.45	1.00	2.01
Dividend Decrease	58,357	0.09	0.00	0.28	2.92	9.53
Net Repurchases (millions)	50,868	72.77	0.03	278.68	5.63	37.09
Repurchase More than 1% Asset Dummy	66,180	0.32	0.00	0.47	0.76	1.58
Panel C: Corporate Control Variables						
Stock Return Volatility	69,806	53.69%	45.07%	31.95%	1.46	5.29
Change in Stock Return Volatility	52,060	−0.32%	−0.82%	21.44%	0.38	8.20
Earnings Volatility	64,522	5.13%	2.65%	6.77%	2.54	10.09
Change in Earnings Volatility	49,510	0.15%	0.00%	3.11%	1.23	32.55
Stock Return	58,086	13.01%	2.97%	65.51%	2.53	12.84
Sales (millions)	76,562	1,848.79	240.81	5,165.31	4.92	29.71
Assets (millions)	76,676	3,592.47	461.28	11,028.17	5.48	36.15
Market leverage	75,963	41.95%	37.68%	28.30%	0.34	1.89
Tangibility	73,822	22.64%	13.89%	23.75%	1.20	3.47
M/B Ratio	75,962	2.86	1.82	4.09	3.45	20.45
ROA	75,340	2.30%	5.59%	19.49%	−2.71	13.01
Effective Tax Rate	76,560	20.98%	30.46%	33.16%	−2.21	15.42
Cash/Interest Expenses	57,879	160.96	4.66	690.86	6.40	46.35
Dividend Yield	76,028	0.01	0.00	0.02	2.70	11.44
Financial Deficit/Sales	39,183	0.49	0.00	2.58	7.11	55.84
Cash Flow/PPE	67,462	−0.51	0.30	6.49	−5.07	35.11
Tobin's Q	75,800	1.94	1.33	1.73	3.28	15.73
Net Working Capital/Assets	59,544	0.07	0.05	0.18	0.10	3.52
R&D/Sales	75,670	19.39%	0.00%	97.24%	7.24	57.57
Firm Age (years)	40,142	7.99	7.00	5.32	0.57	2.80
Sales Growth	57,959	14.43%	7.49%	41.26%	3.40	20.12
Neg. Earn. Dummy	76,553	0.28	0.00	0.45	0.98	1.95
Retained Earnings/Assets	75,181	−0.29	0.06	1.47	−4.23	23.28
Panel D: Macroeconomic Variables						
S&P 500 Return	76,513	9.52%	12.78%	18.33%	−0.83	3.19
Default Spread between Baa and Aaa Bonds	73,057	1.02%	0.95%	0.54%	3.21	14.33
Constant 1-year Maturity Treasury Bill Yield	76,251	2.98%	3.34%	2.29%	0.01	1.37
VIX	76,251	21.21%	21.68%	7.12%	0.99	4.74
Industrial Production Growth	76,676	1.97%	3.98%	9.30%	−2.64	12.04

This table reports summary statistics for text-based variables, firm-level corporate policy variables and control variables, and macro-wide control variables in Panel A, B, C, and D, respectively. All continuous variables, except those in Panel D, are winsorized at the 1% and 99% level. Definitions of these variables are provided in Appendix C. The sample consists of U.S. listed companies that filed 10-K reports over the period of 1994 January and 2015 September.

nature can enhance its flexibility and prediction performance.[18] Second, compared with other machine-learning methods, the LASSO procedure conducts variable selection and dimension reduction techniques, which is well suited for assessing the relative strength of our new text-based variable in predicting corporate policies.[19]

We report LASSO regression results based on two different estimation schemes over the period of 1996 to 2014. The first is the rolling scheme. Thus, we fix the 5-year estimation window size and drop distant observations as recent years are added. For example, we start with estimating a LASSO model of a corporate policy variable during the period of 1996 to 2000 and assess its out-of-sample performance using data on the corporate policy in 2002.[20] We estimate the second model using data from 1997 to 2001 and assess its out-of-sample performance in 2003. Overall, we conduct 13 estimations of LASSO regressions based on different estimation windows and report the total number of times each predictor is retained. Following previous studies, we use R-squared to assess out-of-sample prediction performance and report the average R-squared for 13 LASSO regressions.[21] Different from the rolling scheme, the recursive scheme, our second approach, employs all available data for estimation. For example, we estimate a LASSO model using a 6-year window (e.g., 1996 to 2001) to assess its out-of-sample performance in 2002.

We incorporate the universe of textual variables, corporate control variables, and macro variables defined in Appendix C to predict various corporate policies in the LASSO regressions. Specifically, our predictors include three new textual variables (*Risk Change*, *Rising Risk* and *Resolving Risk*), 13 textual variables based on the previous literature, 26 corporate control variables, and five macro variables. We present results pertaining to change in volatility, textual predictors that have been selected at least once among 13 LASSO estimations, and the top ten most successful predictors, among other corporate control and macro variables, due to space limitation.

5.1. Capital structure

Table 4 reports the results on adjustment in capital structure.[22] We start with OLS regressions to examine the impact of risk change on the book leverage ratio (Panel A), as well as debt and equity adjustments (Panel B). Results in Panel A indicate that rising risk is followed by a significant downward adjustment in leverage. Approximately 12.21% of the median absolute change in leverage can be explained by the presence of a median risk change. More strikingly, a one standard deviation increase in risk change is associated with a drop in the book leverage ratio in a magnitude of 1.5 times of the average annual change in leverage in our sample.[23]

Our evidence is largely consistent with the trade-off theory. The statistical power of our text-based risk measure in predicting leverage adjustments is considerably stronger than that of change in stock return volatility, as none of the coefficient estimates of change in volatility is statistically significant. As noted earlier, a large number of past studies focusing on earnings volatility (e.g., Titman and Wessels, 1988; Kester, 1986) or return volatility (e.g., Welch, 2004; Frank and Goyal, 2009) also document a weak relation between risk and leverage ratio, further highlighting the predictive power of our text-based measure.[24]

The coefficient estimates of firm characteristics, such as leverage, size, ROA, are consistent with past work. Large firms are more likely to increase their leverage (e.g., Frank and Goyal, 2009; Faulkender and Peterson, 2006). In addition, firms with higher equity returns are associated with reducing leverage (e.g., Welch, 2004; Faulkender and Peterson, 2006). The leverage ratio is negatively correlated with future change in leverage, confirming that leverage ratios are mean-reverting (Fama and French, 2002; Baker and Wurgler, 2002; Leary and Roberts, 2005).

We then decompose the change in the book leverage ratio into changes in debt versus equity to separate the effects of risk change on these two components. Panel B of Table 4 reports the results. Rising risk is significantly and negatively correlated with future debt level but not significantly correlated with subsequent changes in equity. This result suggests that the impact on leverage is attributable to the debt channel. In unreported analysis, we further find that the negative effect of rising risk is largely concentrated among large, profitable, and investment grade firms. Since it may be relatively easier for these firms to raise external capital when facing risk shocks, our result suggests that the negative relation between rising risk and firm debt is likely due to demand-side factors. Finally, changes in debt are symmetric in response to rising versus resolving risk, as absolute values of coefficient estimates and significance levels are similar in scale.

Our OLS regression results so far indicate that firms substantially reduce debt and leverage when risk increases, and correspondingly increase debt and leverage ratio when risk resolves. To further assess the predictive power of our new measure, we run in-sample and out-of-sample tests using LASSO regressions. Panel C of Table 4 reports the results.

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[18] Hence, LASSO regression results are less prone to the multiple hypothesis testing issues related to OLS regressions. In particular, there is a concern that interpretations drawn from the OLS regression results may depend on the subset of dependent variable we use.

[19] Compared with ridge regression and elastic net procedure, LASSO regressions tend to completely exclude variables from the model, allowing for sparse selection of predictors.

[20] For out-of-sample tests, we use data on corporate policies in 2002 and data on independent variables in 2001.

[21] $R^2 = 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$, where y is the true value, $\hat{y}$ is the predicted value and $\bar{y}$ is the average value of y.

[22] Note that the number of observations drops substantially from 76,676 in the full sample (Table 3) to 22,311 in the main regressions (Table 4). The reasons are twofold: first, we lose a large number of observations due to lack of corporate information from the previous year. Therefore, we cannot compute the annual change in these variables. In addition, there are many missing values regarding one control variable: *Financial Deficit*.

[23] We acknowledge the possibility that our risk change measure may capture other underlying observable or unobservable factors that can potentially affect corporate decisions. As such, the analysis on economic magnitude of coefficient estimates based on OLS or logit regressions, here and thereafter, should be interpreted with caution.

[24] Bradley et al. (1984) and Friend and Lang (1988) document a negative relation between earnings volatility and leverage. These papers focus on the cross-sectional effects (i.e., investigating the relation of contemporary risk level to leverage ratio), while our paper analyzes changes in leverage after experiencing changing risk.

Table 4
Risk change and capital structure.

Panel A: Book Leverage Ratio Adjustment

	$dlev_{t+1}$		
	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)
Risk Measure	−1.036	−0.003	0.003
	(−2.93)	(−2.36)	(1.78)
Log Sales	0.006	0.006	0.006
	(9.26)	(9.28)	(9.18)
Book Leverage	−0.098	−0.098	−0.098
	(−16.48)	(−16.46)	(−16.48)
Stock Return	−0.002	−0.002	−0.002
	(−2.34)	(−2.30)	(−2.24)
ROA	0.168	0.167	0.168
	(13.59)	(13.57)	(13.78)
Tangibility	0.011	0.011	0.011
	(2.99)	(2.94)	(2.98)
Change in Volatility	−0.004	−0.004	−0.004
	(−0.90)	(−0.92)	(−1.03)
Other Controls	Yes	Yes	Yes
Observations	22,311	22,311	22,311
Adj. R-sq	0.215	0.215	0.214

Panel B: Debt versus Equity Adjustment

	$\%dDebt_{t+1}$			$\%dEquity_{t+1}$		
	Risk Change	Rising Risk	Resolving Risk	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)	(4)	(5)	(6)
Risk Measure	−8.704	−0.020	0.019	−1.310	0.001	0.009
	(−2.44)	(−2.81)	(2.63)	(−0.50)	(0.07)	(1.07)
Log Sales	−0.001	−0.001	−0.001	−0.021	−0.021	−0.021
	(−0.19)	(−0.25)	(−0.24)	(−9.35)	(−9.31)	(−9.23)
Book Leverage	−0.465	−0.465	−0.466	0.034	0.035	0.034
	(−13.00)	(−12.99)	(−12.97)	(0.89)	(0.89)	(0.88)
Stock Return	0.065	0.066	0.066	0.039	0.039	0.039
	(4.55)	(4.59)	(4.63)	(2.68)	(2.68)	(2.66)
ROA	0.073	0.075	0.079	−0.718	−0.716	−0.717
	(1.19)	(1.22)	(1.28)	(−7.66)	(−7.69)	(−7.71)
Tangibility	0.022	0.022	0.023	−0.002	−0.002	−0.002
	(0.83)	(0.81)	(0.84)	(−0.08)	(−0.08)	(−0.07)
Change in Volatility	−0.029	−0.031	−0.033	0.017	0.016	0.016
	(−0.84)	(−0.91)	(−0.97)	(0.68)	(0.65)	(0.67)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,680	22,680	22,680	22,311	22,311	22,311
Adj. R-sq	0.072	0.071	0.071	0.194	0.194	0.194

Panel C: Evidence from LASSO Regressions

	The Rolling Scheme			The Recursive Scheme		
	$dlev_{t+1}$	$\%dDebt_{t+1}$	$\%dEquity_{t+1}$	$dlev_{t+1}$	$\%dDebt_{t+1}$	$\%dEquity_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)
Risk Change	6	11	0	13	13	0
Rising Risk	3	1	0	6	1	0
dLM Hrvd Negative	2	3	0	2	1	0
dFluidity	2	3	0	2	1	0
dLM Positive	0	2	0	0	0	0
dHrvd Positive	0	1	1	0	0	0
dLM Negative	1	0	0	0	0	0
Change in Volatility	0	0	0	1	2	0
Book Leverage	13	13	2	13	13	0
ROA	13	0	13	13	0	13
Retained Earnings/Assets	13	0	13	13	0	13
Log Sales	10	0	11	10	0	13
S&P 500 Return	9	6	1	13	13	0
M/B Ratio	3	0	13	7	0	13

(*continued on next page*)

Table 4 (*continued*).

Panel C: Evidence from LASSO Regressions

	The Rolling Scheme			The Recursive Scheme		
	$dlev_{t+1}$	$\%dDebt_{t+1}$	$\%dEquity_{t+1}$	$dlev_{t+1}$	$\%dDebt_{t+1}$	$\%dEquity_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)
Industrial Production Growth	9	6	2	9	7	0
Market Leverage	5	13	0	1	13	0
Stock Return	0	6	3	3	12	7
Negative Earning	10	2	3	10	2	0
No. of Estimations	13	13	13	13	13	13
Average In-Sample R-sq	0.224	0.079	0.229	0.208	0.092	0.267
Average Out-of-Sample R-sq	0.217	0.034	0.202	0.224	0.048	0.209

This table reports estimation results of the capital structure regressions. Panels A and B report OLS regression results of book leverage ratio adjustment (Panel A) and debt and equity adjustments (Panel B). Panel C reports the in-sample estimation results (the number of times independent variables are retained in LASSO regressions) and out-of-sample results (R-squared for out-of-sample predictions) using LASSO procedure. The sample consists of U.S. listed companies that filed 10-K reports over the period of 1994 January and 2015 September in Panels A and B, and over the period of 1996 and 2014 in Panel C. The dependent variables include active change in book leverage ratio ($dlev_{t+1}$), percentage change in total liabilities ($\%dDebt_{t+1}$) and common equity after adjusting for changes in retained earnings ($\%dEquity_{t+1}$) from year t to $t+1$. Other controls in Panels A and B include M/B ratio, effective tax rate, cash/interest expenses, dividend yield, financial deficit/sales, S&P 500 return, industrial production growth, VIX, default spread, Treasury bill yield, industry, and year fixed effects. The numbers in parentheses are t-statistics with robust standard errors two-way clustered at the firm and year level. Under the rolling scheme in Panel C, we fix a five-year rolling estimation window, and use Lasso procedure to conduct out-of-sample predictions for each year during 2002 to 2014 (13 estimations in total), respectively. The recursive scheme uses all available data in previous years instead. In panel C, we incorporate the universe of textual variables, corporate control variables and macro variables defined in Appendix C as predictors for LASSO regressions. We present results pertaining to change in volatility, textual predictors that have been selected at least once among 13 LASSO estimations, and the top ten most successful predictors among other corporate control and macro variables due to space limitation.

The results in Panel C of Table 4 reveal that our text-based risk measure is among the top five predictors that are retained most frequently by LASSO regressions of capital structure decisions. Consistent with the OLS regression results, our risk measure is particularly successful in predicting percentage changes in debt. It is retained by LASSO for 13 (11) times in such predictions under the recursive (rolling) scheme. In contrast, the conventional measure of change in volatility is selected twice (zero times) under the recurve (rolling) scheme. Among the 13 textual variables that are constructed based on existing dictionaries, only five are retained at least once by the model. The relatively successful predictors are the measure based on the combination of negative words in Loughran–McDonald Sentiment Word Lists and Harvard Psychosociological Dictionary and the measure based on product market fluidity, with both measures being retained at most three times by LASSO procedures under either estimation scheme.

The results show that LASSO models on changes in capital structure have robust in-sample and out-of-sample prediction performance. On average, the in-sample R-squared and out-of-sample R-squared are 22.4% (20.8%) and 21.7% (22.4%), respectively, under the recursive (rolling) scheme. The other four most successful predictors beside our new measure are book leverage, ROA, retained earnings/sales and log sales, largely consistent with our OLS estimation results and the previous literature. Taking together, our results on capital structure decisions confirm that the new text-based measure has strong predictive power when using both OLS and LASSO regressions.

5.2. Investment decisions

In this section, we employ the text-based risk measure to predict adjustments in a wide range of investment decisions including capital expenditure, employment, and R&D decisions. Table 5 reports the results. Panel A presents the OLS regression results on the relation between risk change and adjustments in capital expenditure and employment. We find that rising risk is negatively correlated with future changes in capital expenditure. The reduction in capital expenditure following a median risk change accounts for 16.96% of the median annual percentage change in capital expenditure, while changes in the conventional volatility-based risk measure cannot significantly explain adjustments in capital expenditure. Furthermore, a one standard deviation increase in risk change is associated with a decrease in investment with a magnitude of 9.77% of the sample average. The overall relation between risk change and capital expenditure displays symmetric responses to rising versus resolving risk.

Panel A also examines adjustments of employment decisions, as employment can be regarded as investment in human capital. Experiencing rising risk, firms lower not only their capital expenditure but also employment. A median level of risk change is associated with a 7.77% of median annual change in employment. More strikingly, a one standard deviation increase in risk change is associated with drop in employment in a magnitude of 12.27% of the sample average. In comparison, the volatility-based measure is statistically insignificant. The overall impact of our risk change measure on employment is symmetric for rising versus resolving risk.

To obtain a more complete outlook of corporate investment, we use OLS regressions to examine the impact of risk change on R&D in Panel B. R&D activities are crucial for innovation and long-term growth, although R&D expenses may not yield immediate profitability. R&D projects also exhibit higher probability of failure and their ultimate outcomes are largely uncertain compared with other investment activities. We measure R&D activities by R&D expenses and advertising expenses, respectively. We include

Table 5
Risk change and investment decisions.

Panel A: Capital Expenditure and Employment

	$\%dcapx_{t+1}$			$\%demp_{t+1}$		
	Risk Change	Rising Risk	Resolving Risk	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)	(4)	(5)	(6)
Risk Measure	−17.100	−0.041	0.059	−5.919	−0.013	0.016
	(−3.87)	(−3.88)	(3.28)	(−6.19)	(−4.18)	(4.16)
Log Sales	−0.087	−0.087	−0.087	−0.006	−0.006	−0.006
	(−19.23)	(−19.11)	(−19.22)	(−5.48)	(−5.63)	(−5.49)
Tangibility	−0.558	−0.560	−0.556	−0.037	−0.037	−0.036
	(−8.29)	(−8.31)	(−8.31)	(−3.12)	(−3.19)	(−3.10)
Stock Return	0.283	0.284	0.283	0.041	0.041	0.041
	(15.67)	(15.79)	(15.94)	(9.46)	(9.46)	(9.69)
Tobin's Q	0.021	0.021	0.021	0.015	0.015	0.015
	(2.42)	(2.44)	(2.47)	(8.69)	(8.59)	(8.81)
Change in Volatility	−0.036	−0.040	−0.042	−0.016	−0.017	−0.018
	(−0.75)	(−0.83)	(−0.86)	(−1.71)	(−1.82)	(−1.98)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,592	34,592	34,592	34,542	34,542	34,542
Adj. R-sq	0.076	0.075	0.076	0.089	0.088	0.088

Panel B: R&D and Advertising Expense

	$\%dxrd_{t+1}$			$\%dxad_{t+1}$		
	Risk Change	Rising Risk	Resolving Risk	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)	(4)	(5)	(6)
Risk Measure	−9.429	−0.020	0.029	−13.450	−0.030	0.038
	(−4.37)	(−2.50)	(3.090)	(−4.14)	(−2.66)	(3.99)
Log Sales	−0.019	−0.019	−0.019	−0.021	−0.022	−0.022
	(−6.56)	(−6.65)	(−6.50)	(−9.81)	(−9.54)	(−9.72)
Tangibility	−0.145	−0.147	−0.146	−0.078	−0.079	−0.077
	(−3.96)	(−3.99)	(−3.97)	(−2.46)	(−2.47)	(−2.39)
Stock Return	0.061	0.062	0.062	0.088	0.089	0.089
	(7.31)	(7.36)	(7.59)	(5.81)	(5.80)	(5.84)
Tobin's Q	0.027	0.027	0.027	0.028	0.028	0.028
	(5.73)	(5.70)	(5.78)	(3.17)	(3.18)	(3.20)
Change in Volatility	−0.055	−0.056	−0.057	−0.013	−0.016	−0.019
	(−2.28)	(−2.31)	(−2.38)	(−0.32)	(−0.40)	(−0.46)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,321	15,321	15,321	12,209	12,209	12,209
Adj. R-sq	0.083	0.083	0.083	0.051	0.051	0.051

Panel C: Evidence from LASSO Regressions

	The Rolling Scheme				The Recursive Scheme			
	$\%dcapx_{t+1}$	$\%demp_{t+1}$	$\%dxrd_{t+1}$	$\%dxad_{t+1}$	$\%dcapx_{t+1}$	$\%demp_{t+1}$	$\%dxrd_{t+1}$	$\%dxad_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Risk Change	6	12	11	9	13	13	13	13
Resolving Risk	0	2	6	0	0	1	6	0
dLM Hrvd Negative	5	5	5	0	8	8	6	0
dLM Negative	3	6	3	2	4	3	0	12
dFluidity	0	0	4	0	0	0	9	0
dHrvd Negative	0	0	1	0	0	1	0	0
Change in Volatility	6	2	5	3	1	0	0	4
Tobin's Q	12	13	12	12	13	13	13	12
Stock Return	13	13	13	7	13	13	10	7
Negative Earning	5	13	11	3	13	13	13	5
Market Leverage	10	13	11	0	13	13	11	0
Sales Growth	4	12	7	10	5	12	4	7
ROA	0	4	12	0	0	12	13	0
S&P 500 Return	8	3	0	2	13	13	0	0
Industrial Production Growth	5	7	6	0	7	6	7	0
M/B Ratio	6	0	1	7	13	3	0	7
Dividend Dummy	0	7	2	0	7	10	6	0
No. of Estimations	13	13	13	13	13	13	13	13
Average In-Sample R-sq	0.062	0.072	0.079	0.041	0.067	0.080	0.100	0.043

(*continued on next page*)

Table 5 (*continued*).

Panel C: Evidence from LASSO Regressions								
	The Rolling Scheme				The Recursive Scheme			
	$\%dcapx_{t+1}$	$\%demp_{t+1}$	$\%dxrd_{t+1}$	$\%dxad_{t+1}$	$\%dcapx_{t+1}$	$\%demp_{t+1}$	$\%dxrd_{t+1}$	$\%dxad_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Average Out-of-Sample R-sq	0.034	0.039	0.029	0.012	0.039	0.039	0.024	0.018

This table reports estimation results of the investment decisions. Panels A and B report OLS regression results of capital expenditure and employment adjustment (Panel A) and R&D and advertising adjustments (Panel B). Panel C reports the in-sample estimation results (the number of times independent variables are retained in LASSO regressions) and out-of-sample results (R-squared for out-of-sample predictions) using LASSO procedure. The sample consists of U.S. listed companies that filed 10-K reports over the time period of 1994 January and 2015 September in Panels A and B, and over the period of 1996 to 2014 in Panel C. The dependent variables include percentage change in capital expenditure ($\%dcapx_{t+1}$), employment ($\%demp_{t+1}$), R&D ($\%dxrd_{t+1}$) and advertising expense ($\%dxad_{t+1}$) from year t to t+1. Other controls in Panels A and B include market leverage, cash flow/PPE, ROA, M/B ratio, and industry and year fixed effects. The numbers in parentheses are *t*-statistics with robust standard errors two-way clustered at the firm and year level. In panel C, we incorporate the universe of textual variables, corporate control variables and macro variables defined in Appendix C as predictors for LASSO regressions. We present results pertaining to change in volatility, textual predictors that have been selected at least once among 13 LASSO estimations, and the top ten most successful predictors among other corporate control and macro variables due to space limitation.

advertising expenses because it is shown to be related to product development and R&D investment in previous studies. In addition, data on advertising expenses involve less missing observations.

Our results in Panel B reveal that R&D and advertising expenses significantly decrease upon increasing risk. A median risk change is associated with a 6.87% median change in R&D expenses. Alternatively, a one standard deviation increase in risk change implies a drop in R&D in a magnitude of 30.25% of the sample average. Firms also respond symmetrically to rising versus resolving risk. They increase R&D when risk rises and diminish R&D when risk resolves.

Overall, our evidence from OLS regressions indicates that firms decrease capital expenditure, R&D, and employment following increases in risk, consistent with the theoretical predictions in Bolton et al. (2011, 2013) and Bloom (2009). As discussed earlier, existing theoretical and empirical literature have mixed implications on the relation between uncertainty and corporate investment. By showing that firms decrease their investment and employment activities following increases in down-side risk, our paper significantly adds to this literature.

Panel C of Table 5 reports LASSO regression results for investment decisions. The results reveal that proxies indicating investment or growth opportunities, including Tobin's Q and stock return, are the most successful predictors for corporate investment. On average, Tobin's Q and stock return are retained 12 and 11 times, respectively, by LASSO regressions across different types of investment decisions and estimation schemes. Our text-based measure has strong predictive power to a similar extent: the new measure on average is selected 11 times by LASSO regressions. In contrast, the textual variable developed based on the combination of negative words in Loughran–McDonald Sentiment Word Lists and Harvard Psychosociological Dictionary is selected only four times by the model, on average, while the remaining textual variables and the change in volatility measure have an even lower likelihood of being selected. Our results confirm that the new risk measure substantially outperforms existing textual and non-textual risk measures in predicting corporate investment activities.

5.3. *Cash holdings*

In this section, we test whether our new text-based measure can predict firms' adjustment in cash holdings. Panel A of Table 6 reports the OLS regression results. The results indicate that rising downside risk is positively associated with changes in corporate cash holdings. A one standard deviation increase in risk change is associated with an increase in cash-to-asset ratio in a magnitude of 63.24% of the sample average. Meanwhile, a median risk change explains almost 98% of the median annual change in cash-to-asset ratio during the sample period. Our evidence is consistent with Opler et al. (1999) and Bates et al. (2009), who show that the ratio of cash to total non-cash assets is higher for firms with riskier cash flows.

Implications of change in volatility for cash holdings are smaller even when the relation between change in volatility and cash holdings is statistically significant in OLS regressions. To illustrate, a median risk change explains 28.39% of the median change in the cash-to-asset ratio. Consistent with previous studies, large firms with more net working capital tend to increase their cash holdings. In contrast, firms with higher level of cash or with large increase in cash-to-asset ratio are more likely to reduce subsequent cash holdings. Analyzing rising versus resolving risk reveals symmetric responses. Increasing risk corresponds to increasing cash holdings and vice versa.

Panel B reports findings from LASSO regressions. The evidence indicates that prior cash-to-asset ratio and prior changes in cash-to-asset ratio have the strongest predictive power: these two predictors are both retained 13 times by LASSO procedures under either rolling or recursive schemes. Our text-based measure is the next most successful predictor with the variable being selected nine and 13 times, respectively, per the rolling and recursive schemes. Different than previous corporate policies examined, change in volatility has some power in predicting cash holdings. The variable is retained five and eight times, respectively, under the rolling and recursive schemes. Textual measures based on product market fluidity and various positive or negative word dictionaries also are selected at least once among 13 Lasso regressions. These measures, however, substantially underperform our proposed text-based measure.

Table 6
Risk change and cash holdings.

Panel A: OLS regression results

	$dCash_{t+1}$		
	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)
Risk Measure	1.548	0.004	−0.004
	(5.750)	(3.150)	(−3.17)
Log Sales	0.003	0.003	0.003
	(4.910)	(4.930)	(4.830)
Net Working Capital/Assets	0.053	0.053	0.053
	(10.360)	(10.390)	(10.450)
R&D/Sales	−0.001	−0.001	−0.001
	(−2.10)	(−2.08)	(−2.17)
Dividend Dummy	−0.002	−0.002	−0.002
	(−1.71)	(−1.64)	(−1.83)
$dCash_t$	−0.165	−0.165	−0.165
	(−13.94)	(−13.92)	(−13.80)
Cash	−4.670	−4.660	−4.680
	(−5.40)	(−5.44)	(−5.45)
Change in Volatility	0.010	0.010	0.010
	(2.210)	(2.300)	(2.410)
Other Controls	Yes	Yes	Yes
Observations	30,456	30,456	30,456
Adj. R-sq	0.064	0.063	0.063

Panel B: Evidence from LASSO Regression

	The Rolling Scheme	The Recursive Scheme
	$dCash_{t+1}$	$dCash_{t+1}$
	(1)	(2)
Risk Change	9	13
dFluidity	3	6
dHrvd Positive	4	4
dLM Negative	2	6
dHrvd Negative	3	4
dLM Hrvd Negative	1	1
dLM Positive	1	0
Change in Volatility	5	8
Cash/Assets	13	13
$dCash_t$	13	13
Cash	9	10
S&P 500 Return	9	10
Log Sales	6	11
R&D/Sales	4	10
Retained Earnings/Asset	4	10
Book Leverage	2	11
Sales Growth	2	10
Industrial Production Growth	6	5
Tangibility	1	7
No. of Estimations	13	13
Average In-Sample R-sq	0.054	0.052
Average Out-of-Sample R-sq	0.021	0.020

This table reports estimation results from cash holdings regressions. Panel A reports OLS regression results, and Panel B reports the in-sample estimation results (the number of times independent variables are retained in LASSO regressions) and out-of-sample results (R-squared for out-of-sample predictions) using LASSO procedure. The sample consists of U.S. listed companies that filed 10-K reports over the period of 1994 January and 2015 September in Panel A, and over the period of 1994 t0 2014 in Panel B. The dependent variable $dCash_{t+1}$ denotes change in cash/assets from year t to $t+1$. Other controls in Panel A include M/B ratio, cash flow/PPE, capital expenditure/assets, book leverage, industry, and year fixed effects. The numbers in parentheses are t-statistics with robust standard errors two-way clustered at the firm and year level. In panel B, we incorporate the universe of textual variables, corporate control variables and macro variables defined in Appendix C as predictors for LASSO regressions. We present results pertaining to change in volatility, textual predictors that have been selected at least once among 13 LASSO estimations, and the top ten most successful predictors among other corporate control and macro variables due to space limitation.

5.4. Payout policy

In this section, we use our new downside risk measure to comprehensively examine payout policies, including dividend initiations, omissions, increases and decreases as well as share repurchases.

Table 7 reports the results on dividends, while Table 8 focuses on repurchase decisions. Panel A of Table 7 presents logistic regression results where the dependent variable is a dummy variable reflecting dividend initiation, dividend omission, dividend increase, and dividend decrease, respectively. The evidence shows that increases in risk are positively and significantly correlated with dividend omission and dividend decrease, and negatively correlated with dividend initiation and dividend increase. As risk change increases by one standard deviation above the sample mean, there are 12.36% and 10.34% increases in the likelihood of dividend omission and dividend decrease, respectively. Such increases in risk change are also associated with 12.86% and 4.62% decreases in the likelihood of dividend initiation and dividend increase, respectively.

It should be noted that change in volatility is also statistically significant in predicting dividend policies based on logit regressions, and it is of similar economic magnitude to our text-based measure. For example, increasing the annual stock return volatility by one standard deviation above the sample mean leads to a 15.46% decrease in the propensity of dividend initiation.

We also use logit regressions to examine whether rising and resolving risk affect dividend policy in the same manner. Panels B presents the results. The evidence shows that firms respond to rising risk by substantially adjusting dividend policy along multiple dimensions. In particular, dividend-paying firms are less likely to increase dividends. Some of them may choose to moderately reduce dividend payouts, while others dramatically terminate dividend payouts. On the other hand, when risk resolves, dividend-paying firms may not immediately increase dividends.

Panel C of Table 7 reports LASSO estimation results. We find that LASSO models on dividend increase yield much better in-sample and out-of-sample prediction performance than estimations of dividend initiation, dividend decrease and dividend omission. The in-sample R-Squared and out-of-sample R-squared are 47% (50%) and 51% (52%), respectively, under the recursive (rolling) scheme for predicting dividend increases. There are only seven predictors that are retained at least once by LASSO among 13 such predictions. The most successful ones include our new text-based measure and the indicators of whether firms issue dividends or whether firms have negative earnings in the previous year, with all three predictors being retained consistently for 13 times under the recursive scheme. The remaining four predictors (i.e., market leverage, cash holdings, and textual measures based on product market fluidity and negative words) have significantly lower predictive power than the above top three predictors. Taking together, our evidence from logit and LASSO regressions is supportive of the notion that firms adjust their dividend policies in response to changes in perceived risk (e.g., Hoberg and Prabhala, 2009; Hoberg et al., 2014).

Share repurchases is as an alternative way to pay dividends. Jagannathan et al. (2000) and Guay and Harford (2000) find that repurchases is used to distribute cash flow shocks that are primarily transient, while dividend changes typically follow cash flow shocks with a relatively large permanent component. Table 8 reports our results on stock repurchases.

The OLS and logit regressions results in Panel A reveal similar findings to those on dividend payouts. First, we show that increases in risk are followed by a significant reduction in the propensity of large stock repurchases. Particularly, a one standard deviation increase in the risk change corresponds to an 8.94% decrease in the probability of large stock repurchases. Moreover, the net value of repurchase is negatively related to rising risk. Net stock repurchases decreases in the magnitude of 0.39 million following a median risk change, while the median change in the net repurchases is only 0.03 million in the full sample. In contrast, change in stock return volatility does not significantly predict either net repurchases or the likelihood of large repurchases. Our results also reveal that firms adjust repurchase policies upon facing rising risk but may not reverse these policies when risk resolves.

The LASSO estimation results in Panel B reveal that firm size, cash holdings, investment opportunities, as captured by Tobin's Q and market-to-book ratio, market leverage, and dividend dummy are the most successful predictors of repurchase policies. Although our text-based measure has lower predictive power than these top predictors, it is much more likely to be selected by LASSO than change in volatility, existing text-based measures and the remaining macro and corporate variables.

In sum, our empirical results on cash holdings and payout policies are largely consistent with the idea that an increase in risk increases the precautionary demand for cash. Thus, managers, being characterized as conservative in field surveys Lintner, 1956; Brav et al., 2005, would reduce payouts and retain cash for future investment.

5.5. Extending predictability to the longer run

Our results so far indicate that the new text-based measure is predictive of corporate policies in the following year. We further examine whether the new measure can predict corporate policies in the longer term. We include lagged terms of risk change measured at time t, $t-1$, and $t-2$ while keeping the control variables observed at time t in the OLS regressions specified below

$$\triangle \mathrm{POLICY}_{i,t+1} = \alpha + \beta_1 \mathrm{RISKCHANGE}_{i,t} + \beta_2 \mathrm{RISKCHANGE}_{i,t-1} + \beta_3 \mathrm{RISKCHANGE}_{i,t-2} + \beta_j \mathrm{CONTROL}_{i,j,t} + \varepsilon_{i,t} \tag{2}$$

Table 9 reports the results. Panels A through E present the results on leverage, investment, cash holdings, dividend payouts, and stock repurchases, respectively. The evidence shows that our risk change measure can predict changes in payout policies including dividends and repurchases persistently two years after an initial increase in risk. Furthermore, the risk change measure can predict changes in leverage, investment, and cash holdings persistently three years following an increase in risk.

Taking together the findings exhibited in Table 9, we show that our new measure can predict adjustments in corporate policies in the longer run, particularly for investment and capital structure policies. One interpretation could be that the new measure captures deteriorating long-term business prospects as perceived by the management, rather than short-lasting investor sentiment or biases in investors' expectations. This new measure is, therefore, more capable of predicting persistent adjustment in corporate decisions than existing measures including changes in stock return volatility and earnings volatility.

Table 7
Risk change and dividend policy.

Panel A: The Impact of Risk Chang

	Dividend $Initiation_{t+1}$	Dividend $Omission_{t+1}$	Dividend $Increase_{t+1}$	Dividend $Decrease_{t+1}$
	(1)	(2)	(3)	(4)
Risk Change	−86.590	72.790	−31.090	62.400
	(−2.02)	(2.41)	(−1.81)	(2.42)
Log Firm Age	−0.103	0.147	0.096	0.129
	(−0.49)	(1.19)	(1.29)	(1.73)
Sales Growth	0.058	−0.900	−0.118	−0.618
	(0.18)	(−1.60)	(−0.72)	(−1.92)
Neg. Earn.	−0.937	1.216	−1.444	0.608
Dummy	(−3.24)	(5.32)	(−11.06)	(4.21)
R&D/Sales	−2.031	−7.928	−6.229	−8.217
	(−1.48)	(−4.84)	(−2.90)	(−3.77)
Retained	0.087	0.372	0.605	0.415
Earnings/Assets	(0.65)	(1.78)	(2.64)	(3.63)
M/B Ratio	−0.018	−0.004	−0.001	−0.020
	(−0.60)	(−0.11)	(−0.05)	(−1.02)
ROA	2.080	0.654	2.053	1.470
	(1.45)	(0.55)	(1.84)	(1.92)
Log Sales	0.001	−0.028	0.282	0.057
	(0.02)	(−0.38)	(6.96)	(1.38)
Change in	−1.022	1.859	−0.418	1.488
Volatility	(−2.19)	(8.81)	(−1.72)	(6.44)
Other Controls	Yes	Yes	Yes	Yes
Observations	18,189	17,946	18,633	18,479

Panel B: Rising versus Resolving Risk

	Dividend $Initiation_{t+1}$		Dividend $Omission_{t+1}$		Dividend $Increase_{t+1}$		Dividend $Decrease_{t+1}$	
	Rising (1)	Resolving (2)	Rising (3)	Resolving (4)	Rising (5)	Resolving (6)	Rising (7)	Resolving (8)
Risk Measure	−0.147	0.134	0.306	0.017	−0.190	−0.133	0.216	0.007
	(−1.04)	(1.46)	(2.43)	(0.12)	(−3.14)	(−0.48)	(2.190)	(0.120)
Log Firm Age	0.002	0.003	0.062	0.063	0.155	0.151	0.057	0.059
	(0.03)	(0.04)	(0.73)	(0.73)	(2.96)	(2.88)	(0.74)	(0.77)
Sales Growth	−0.436	−0.438	−1.274	−1.310	−0.289	−0.276	−0.720	−0.735
	(−1.91)	(−1.94)	(−2.35)	(−2.35)	(−3.00)	(−2.91)	(−4.06)	(−4.18)
Neg. Earn.	−0.653	−0.671	0.872	0.969	−1.272	−1.344	0.315	0.383
Dummy	(−2.70)	(−2.86)	(4.76)	(5.08)	(−9.41)	(−10.02)	(2.83)	(3.11)
R&D/Sales	−0.034	−0.027	−0.953	−0.982	−2.037	−2.020	−3.312	−3.338
	(−0.15)	(−0.13)	(−1.62)	(−1.63)	(−0.90)	(−0.89)	(−1.88)	(−1.89)
Retained	0.127	0.127	0.145	0.153	0.570	0.549	0.198	0.209
Earnings/Assets	(1.24)	(1.25)	(1.35)	(1.39)	(3.42)	(3.34)	(1.51)	(1.56)
M/B Ratio	−0.028	−0.027	−0.011	−0.011	−0.008	−0.008	−0.016	−0.016
	(−1.68)	(−1.63)	(−0.37)	(−0.39)	(−0.79)	(−0.73)	(−1.44)	(−1.46)
ROA	2.067	2.096	1.199	1.229	2.362	2.365	1.986	1.976
	(4.14)	(4.14)	(1.47)	(1.50)	(3.63)	(3.67)	(3.22)	(3.20)
Log Sales	−0.020	−0.020	−0.146	−0.136	0.261	0.258	0.031	0.036
	(−0.72)	(−0.73)	(−3.46)	(−3.04)	(8.83)	(8.72)	(1.01)	(1.13)
Change in	−1.033	−1.034	1.855	1.878	−0.397	−0.427	1.488	1.514
Volatility	(−2.19)	(−2.18)	(8.76)	(8.87)	(−1.70)	(−1.78)	(6.39)	(6.52)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,189	18,189	17,946	17,946	18,633	18,633	18,479	18,479

Panel C: PEvidence from LASSO Regressions

	The Rolling Scheme				The Recursive Scheme			
	Dividend $Initiation_{t+1}$ (1)	Dividend $Omission_{t+1}$ (2)	Dividend $Increase_{t+1}$ (3)	Dividend $Decrease_{t+1}$ (4)	Dividend $Initiation_{t+1}$ (5)	Dividend $Omission_{t+1}$ (6)	Dividend $Increase_{t+1}$ (7)	Dividend $Decrease_{t+1}$ (8)
Risk Change	4	4	9	9	2	5	13	9
dFluid	0	7	1	4	0	8	1	2
dLM Negative	0	3	0	3	0	4	0	5
dLM Hrvd Negative	0	3	1	1	1	0	1	1

(*continued on next page*)

Table 7 (*continued*).

Panel C: PEvidence from LASSO Regressions

	The Rolling Scheme				The Recursive Scheme			
	Dividend Initiation$_{t+1}$ (1)	Dividend Omission$_{t+1}$ (2)	Dividend Increase$_{t+1}$ (3)	Dividend Decrease$_{t+1}$ (4)	Dividend Initiation$_{t+1}$ (5)	Dividend Omission$_{t+1}$ (6)	Dividend Increase$_{t+1}$ (7)	Dividend Decrease$_{t+1}$ (8)
dHrvd Positive	1	0	0	1	5	0	0	1
Change in Volatility	0	6	0	6	3	7	0	7
Dividend Dummy	13	7	13	13	13	7	13	13
Market Leverage	0	13	3	8	0	13	8	7
Dividend Yield	10	0	0	7	10	0	0	7
Neg. Earn. Dummy	0	0	11	1	0	0	13	7
S&P 500 Return	6	1	2	2	9	1	0	0
Log Firm Age	1	0	0	2	0	7	0	7
Industrial Production Growth	3	0	0	0	0	0	0	0
dCash$_t$	0	1	0	1	0	0	0	0
Cash	0	0	0	0	0	0	2	0
No. of Estimations	13	13	13	13	13	13	13	13
Average In-Sample R-sq	0.096	0.017	0.500	0.070	0.086	0.012	0.470	0.080
Average Out-of-Sample R-sq	0.039	0.001	0.520	0.020	0.057	0.001	0.510	0.030

This table reports the estimation results on dividend policy. Logistic regression results are presented with the main independent variable being *Risk Change* from year $t-1$ to t in Panel A and *Rising Risk* and *Resolving Risk* from year $t-1$ to t in Panel B, respectively. Panel C reports the in-sample estimation results (the number of times independent variables are retained in LASSO regressions) and out-of-sample results (R-squared for out-of-sample predictions) using LASSO procedure. The sample consists of U.S. listed companies that filed 10-K reports over the time period of 1994 January and 2015 September in Panels A and B, and over the period of 1996 to 2014 in Panel C. The main dependent variables are dividend initiation, dividend omission, dividend increase, and dividend decrease at year $t+1$. *Dividend initiation* (*Dividend Omission*) is an indicator that equals one if the company initiates (omits) dividends in a certain year. *Dividend Increase* (*Dividend Decrease*) is an indicator that equals one if the company increases (decreases) dividends in a certain year. Other controls in Panels A and B include industry and year fixed effects. The numbers in parentheses are t-statistics with robust standard errors two-way clustered at the firm and year level. In panel C, we incorporate the universe of textual variables, corporate control variables and macro variables defined in Appendix C as predictors for LASSO regressions. We present results pertaining to change in volatility, textual predictors that have been selected at least once among 13 LASSO estimations, and the top ten most successful predictors among other corporate control and macro variables due to space limitation.

6. Robustness checks

This section implements a comprehensive battery of robustness checks. First and foremost, we examine alternative downside risk measures based on different sections of 10-K texts. In particular, Item 7 and 7a (MD&A section) in 10-K filings contain comprehensive managerial discussions on corporate risk profile and business prospects. Using a risk change measure derived solely from Item 7 and 7a texts, our results remain unchanged. Moreover, Hoberg et al. (2014) find that product market risk information extracted from Item 1 (Business) can predict future cash holdings and payout policies. Here, we find that a risk change measure based on Item 1 yields similar results as our benchmark measures. Next, since listed companies are mandated by SEC to include Item 1a to discuss “the most significant factors that make the company speculative or risky” after December 2005, we analyze risk information embedded in Item 1a separately.[25] Our overall evidence remains unchanged, supporting the finding in Campbell et al. (2014) that risk disclosure in Item 1a is firm-specific, meaningful, and relevant to different types of risks, both systematic and idiosyncratic.

Our results are robust to alternative weighting schemes for our risk-related keywords. There is a concern that the word ‘failure’ has a much more negative meaning than other words such as “competition”. To address this concern, we adopt different approaches of imposing heavier weights on keywords related to “failure” and find similar results.

One possible concern is that our main results are attributable to financial firms since the 15 extreme risk change examples are concentrated in the finance industry. We conduct subsample analyses on different industry groups and obtain similar results when restricting our sample to nonfinancial firms or manufacturing firms. Interestingly, financial firms are more responsive to risk change through reduction in leverage and are non-responsive through adjustments in cash holdings. Furthermore, unlike Floyd et al. (2015), we find that financial firms are more likely to terminate their dividends following risk changes than nonfinancial firms. In comparison, manufacturing firms and firms in the service industry cut capital expenditure and R&D to a greater degree following risk changes. Our results are also robust to excluding the 2007–2008 period of the global financial crisis, during which firms reduce leverage to a greater extent than the rest of the sample period.

Our results are robust to using market leverage and adjusted book and market leverages as proxies for capital structure. We also derive similar results using the absolute change in capital expenditure and employment, rather than the percentage change in these measures, as dependent variables. Regarding payout policies, we further investigate the predictability of percentage increase or decrease in dividend payouts and repurchases using our text-based measure. We find similar results upon using these continuous measures, rather than the dummy variables.

[25] One may argue that some sections in the 10-K, even those in Item 7, may be used to describe what has happened in the past. The downside risk measures developed based on these sections, therefore, may not be as forward-looking. Only including disclosure on Item 1a can alleviate such concerns since the entire section is limited to forward-looking discussions on risk factors.

Table 8
Risk change and stock repurchases.

Panel A: OLS and Logistic Regression Results

	Net Repurchases$_{t+1}$(OLS)			Repurchase More than 1% Asset Dummy$_{t+1}$(LOGIT)		
	Risk Change	Rising Risk	Resolving Risk	Risk Change	Rising Risk	Resolving Risk
	(1)	(2)	(3)	(4)	(5)	(6)
Risk Measure	−2221.500	−11.990	9.902	−18.180	−0.090	0.089
	(−3.56)	(−3.57)	(2.99)	(−1.94)	(−2.26)	(2.40)
Log Sales	55.620	55.680	55.640	0.238	0.239	0.237
	(6.440)	(6.46)	(6.44)	(8.48)	(8.51)	(8.30)
Sale Growth	−12.350	−12.380	−12.340	−0.507	−0.506	−0.500
	(−3.85)	(−3.87)	(−3.88)	(−5.48)	(−5.51)	(−5.44)
R&D/Sales	24.670	24.640	24.680	0.127	0.127	0.127
	(5.45)	(5.47)	(5.46)	(2.67)	(2.66)	(2.70)
Retained Earnings/Assets	−8.669	−8.625	−8.635	0.114	0.114	0.111
	(−2.66)	(−2.63)	(−2.64)	(2.88)	(2.87)	(2.82)
Log Firm Age	6.298	6.288	6.267	0.211	0.211	0.210
	(1.63)	(1.63)	(1.62)	(5.94)	(5.89)	(5.89)
M/B Ratio	2.477	2.457	2.463	0.041	0.041	0.042
	(3.10)	(3.08)	(3.09)	(4.44)	(4.41)	(4.47)
ROA	−38.680	−38.610	−38.650	2.140	2.135	2.132
	(−1.49)	(−1.49)	(−1.49)	(5.66)	(5.64)	(5.68)
Neg. Earn. Dummy	−15.760	−14.730	−15.400	−0.518	−0.512	−0.542
	(−4.15)	(−3.84)	(−4.20)	(−7.34)	(−7.12)	(−8.09)
Change in Volatility	7.180	7.702	7.476	−0.399	−0.396	−0.409
	(0.630)	(0.69)	(0.66)	(−1.57)	(−1.56)	(−1.60)
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,577	16,577	16,577	16,627	16,627	16,627

Panel B: Evidence from LASSO Regressions

	The Rolling Scheme		The Recursive Scheme	
	Net Repurchases$_{t+1}$	Repurchase More than 1% Asset Dummy$_{t+1}$	Net Repurchases$_{t+1}$	Repurchase More than 1% Asset Dummy$_{t+1}$
	(1)	(2)	(3)	(4)
Risk Change	8	6	10	6
dFluid	3	2	3	1
dHrvd Negative	1	2	2	3
dLM Hrvd Negative	1	1	0	0
dLM Litigation	2	0	0	0
Change in Volatility	0	0	0	1
Log Sales	13	13	13	13
Cash	13	13	13	13
Tobin's Q	12	13	13	11
Market Leverage	11	12	13	12
Dividend Dummy	8	12	11	10
M/B Ratio	6	12	7	12
Industrial Production Growth	4	4	7	5
S&P 500 Return	3	0	2	1
Log Firm Age	0	0	2	1
No. of Estimations	13	13	13	13
Average In-Sample R-sq	0.330	0.113	0.320	0.094
Average Out-of-Sample R-sq	0.310	0.096	0.320	0.104

This table reports the estimation results on stock repurchases. Panel A reports OLS (Columns (1)–(3)) and Logistic (Columns (4)–(6)) regression results, and Panel B reports the in-sample estimation results (the number of times independent variables are retained in LASSO regressions) and out-of-sample results (R-squared for out-of-sample predictions) using LASSO procedure. The sample consists of U.S. listed companies that filed 10-K reports over the time period of 1994 January and 2015 September in Panel A, and over the period of 1996 to 2014 in Panel B. The main dependent variables include *Net Repurchases* and *Repurchase More than 1% Asset Dummy*, all measured at year $t+1$. Other controls in Panel A include industry fixed effects (industry dummies by the first two-digit SIC code), and year fixed effects. The numbers in parentheses are t-statistics with robust standard errors two-way clustered at the firm and year level. In panel B, we incorporate the universe of textual variables, corporate control variables and macro variables defined in Appendix C as predictors for LASSO regressions. We present results pertaining to change in volatility, and textual predictors and predictors among other corporate control and macro variables that have been selected at least once among 13 LASSO estimations due to space limitation.

We also examine potential effects of CEO traits on predicting corporate policies. Graham et al. (2013) document that CEOs and CFOs around the world possess different personal traits such as risk aversion and optimism, which could affect corporate leverage and

Table 9
Predictability in the long run.

Row	Risk Change (t)	Risk Change ($t-1$)	Risk Change ($t-2$)	Controls	Observations
Panel A: Leverage Adjustment					
Dependent Variable: dlev$_{t+1}$					
(1)	−1.036			Yes	22,311
	(−2.93)				
(2)	−0.864	−0.373		Yes	17,137
	(−2.19)	(−2.96)			
(3)	−1.067	−0.184	−1.209	Yes	13,168
	(−2.77)	(−2.37)	(−2.45)		
Panel B: Investment Adjustment					
Dependent Variable: %dcapx$_{t+1}$					
(4)	−17.110			Yes	34,592
	(−3.87)				
(5)	−14.690	−6.062		Yes	26,291
	(−3.07)	(−3.33)			
(6)	−13.980	−4.754	−1.228	Yes	20,002
	(−2.41)	(−2.96)	(−2.34)		
Dependent Variable: %demp$_{t+1}$					
(7)	−5.918			Yes	34,542
	(−6.19)				
(8)	−4.828	−3.988		Yes	26,279
	(−4.88)	(−6.10)			
(9)	−5.313	−4.579	−3.124	Yes	19,987
	(−5.07)	(−4.93)	(−3.25)		
Dependent Variable: %dxrd$_{t+1}$					
(7)	−9.428			Yes	15,321
	(−4.36)				
(8)	−7.511	−4.542		Yes	11,595
	(−3.25)	(−2.03)			
(9)	−5.047	−5.315	−3.552	Yes	8,793
	(−2.16)	(−1.86)	(−1.81)		
Panel C: Cash Holdings Adjustment					
Dependent Variable: dcash$_{t+1}$					
(10)	1.549			Yes	30,456
	(5.75)				
(11)	1.052	0.805		Yes	24,579
	(3.01)	(1.92)			
(12)	1.454	0.755	1.035	Yes	19,098
	(4.23)	(1.75)	(4.33)		
Panel D: Dividend Policy Adjustment					
Dependent Variable: Dividend Initiation$_{t+1}$*(Logistic Model)*					
(13)	−86.590			Yes	18,189
	(−2.02)				
(14)	−84.650	15.990		Yes	13,248
	(−1.77)	(0.47)			
(15)	−72.340	34.140	10.930	Yes	9,426
	(−1.16)	(1.00)	(0.25)		
Dependent Variable: Dividend Omission$_{t+1}$*(Logistic Model)*					
(16)	72.790			Yes	17,946
	(2.41)				
(17)	79.150	74.270		Yes	12,952
	(2.30)	(1.45)			
(18)	53.850	53.730	−67.330	Yes	8,735
	(1.11)	(0.79)	(−1.18)		
Dependent Variable: Dividend Increase$_{t+1}$*(Logistic Model)*					
(19)	−31.090			Yes	18,633
	(−1.81)				
(20)	−25.700	−21.490		Yes	13,517
	(−1.18)	(−0.99)			
(21)	−8.480	−2.818	17.780	Yes	9,823
	(−0.37)	(−0.11)	(0.79)		
Panel D: Dividend Policy Adjustment					
Dependent Variable: Dividend Decrease$_{t+1}$*(Logistic Model)*					
(22)	62.400			Yes	18,479

(continued on next page)

Table 9 (*continued*).

Row	Risk Change (t)	Risk Change ($t-1$)	Risk Change ($t-2$)	Controls	Observations
	(2.79)				
(23)	60.120	12.050		Yes	13,384
	(2.31)	(1.48)			
(24)	62.270	12.330	11.770	Yes	9,705
	(2.04)	(2.36)	(1.04)		
Panel E: Repurchase Policy Adjustment					
Dependent Variable: Net Repurchase$_{t+1}$					
(25)	−2221.500			Yes	16,577
	(−3.56)				
(26)	−2868.300	−1687.900		Yes	12,177
	(−3.15)	(−3.40)			
(27)	−3531.300	−1692.500	−508.700	Yes	8,952
	(−3.72)	(−2.19)	(−0.58)		

This table examines the predictability of corporate policies in the longer run. The sample consists of U.S. listed companies that filed 10-K reports over the period of 1994 January and 2015 September. We examine change in leverage in Panel A, investment in Panel B, cash holdings in Panel C, and dividend and repurchase policy in Panel D and E, all measured at year $t+1$. The key independent variables include risk change at year t, at year $t-1$, and at year $t-2$. We control for the same set of variables as in the benchmark regressions. In addition, we control for change in volatility at year t, year $t-1$, and year $t-2$ in corresponding regressions. The numbers in parentheses are t-statistics with robust standard errors two-way clustered at the firm and year level.

Table A.1
List of downside risk related keywords.

Key Word Stem	Key Words
accid	accident accidents
advers	adverse adversely adversity adversities
compet	compete competent competing competes competencies competence competency competed competently
competi	competition competitions
competit	competitive competitiveness competitively
competitor	competitors competitor
crisis	crisis
difficult	difficult
difficulti	difficulties difficulty
downgrad	downgrade downgraded downgrades downgrading
downturn	downturn downturns
downward	downward
fail	fail fails failed failing
failur	failure failures
impair	impairment impaired impair impairments impairing impairs
inconsist	inconsistent inconsistency inconsistencies inconsistently
lose	lose losing loses
loss	loss losses
lost	lost
neg	negative negatively negatives
nonperform	nonperforming nonperformance
pressur	pressure pressures pressured pressurized pressurization pressuring pressurizer pressurize
risk	risk risks risked risking
riski	risky riskiness
slowdown	slowdown slowdowns
unabl	unable
weaken	weakening weakened weaken weakens
weaker	weaker
weak	weaknesses weakness

This table present the list of key word stems and corresponding key words for constructing the downside risk measure.

investment decisions.[26] We thus add predictors capturing CEO gender, age, education, and experience. We measure CEO education from a variety of dimensions, including whether the CEO has a bachelor, master, or PhD degree, whether the CEO graduates from an Ivy-league college, whether the CEO obtains an MBA degree, whether the CEO obtains an MBA degree from the "US News (2010)" top 20 MBA programs and whether the CEO obtains a Master of Finance degree. We measure CEO prior experience by examining the number of years the CEO has worked in the finance industry, the number of years the CEO has worked for the same industry as

[26] Graham et al. (2013) find that younger and male CEOs, and CEOs with MBA degrees and longer financial industry experience may adopt more aggressive financial policies.

Table A.2
Examples with downside risk related key word stems.

Key Word Stem	Sentences
accid*	While we believe that we are currently in compliance with all material rules and regulations governing the use of hazardous materials and, to date, we have not had any adverse experiences, in the event of an **accident** or environmental discharge, we may be held liable for any resulting damages, which may exceed our financial resources and may materially harm our business, financial condition and results of operations. (AVALON PHARMACEUTICALS INC, 2008/03/31)
advers*	We cannot assure you that future compliance with these regulations, future environmental liabilities, the cost of defending environmental claims, conducting any environmental remediation, or generally resolving liabilities caused by us or related third parties will not have a material **adverse** effect on our business, financial condition or results of operations. (AMERCO, 2006/06/13)
compet*	Our inability to **compete** effectively against these companies or to maintain our relationships with the various automobile dealers through whom we offer consumer loans could limit the growth of our consumer loan business. Economic and credit market downturns could reduce the ability of our customers to repay loans, which could cause our consumer loan portfolio losses to increase. (FRANKLIN RESOURCES INC, 2002/12/18)
competi*	Our profitability is therefore affected by the prices of lumber which may fluctuate based on several factors beyond our control, including, among others, changes in supply and demand, general economic conditions, labor costs, **competition** and, in some cases, government regulation. (WII Components, Inc., 2007/03/28)
competit*	Changes in industry conditions and the **competitive** environment may impact the accuracy of the Company's projections. (ALPINE GROUP INC, 2002/04/04)
competitor*	The determination as to when in the product development cycle technological feasibility has been established, and the expected product life, require judgments and estimates by management and can be affected by technological developments, innovations by **competitors** and changes in market conditions affecting demand. (HURCO COMPANIES INC, 2005/01/21)
crisis*	The current credit **crisis** may also have a potential impact on our need to obtain additional financing in the future and may impact the determination of fair values, financial instrument classification, or require impairments in the future. (IDM PHARMA, INC, 2009/03/31)
difficult*	Although the acquisition of Carlisle Power Transmission was accretive in 2001, earnings in this segment fell due to extremely **difficult** market conditions and much lower production levels. Also contributing to the lower earnings was intense competitive pressure requiring price concessions to maintain market share. (CARLISLE COMPANIES INC, 2002/03/21)
difficulti*	Thus, our chances of financial and operational success should be evaluated in light of the risks, uncertainties, expenses, delays and **difficulties** associated with operating a business in a relatively new and unproven market or a new business in an existing market, many of which may be beyond our control. (GLOBAL SPORTS INC, 2002/04/04)
downgrad*	If uncertainties in these markets continue, these markets deteriorate further or the Company experiences any additional ratings **downgrades** on any investments in its portfolio (including on ARS), the Company may incur additional impairments to its investment portfolio, which could negatively affect the Company's financial condition, cash flow and reported earnings. (BRISTOL MYERS SQUIBB CO, 2008/02/22)
downturn*	Furthermore, customers who benefit from shorter lead times may defer some purchases to future periods, which could affect our demand and revenues for the short term. As a result, we may experience **downturns** or fluctuations in demand in the future and experience adverse effects on our operating results and financial condition. (LSI LOGIC CORP, 2001/03/08)
downward*	Tunneling gross margins continued to suffer in 2005 from the continuation of large projects that experienced **downward** gross margin revisions. (INSITUFORM TECHNOLOGIES INC, 2007/02/23)
fail*	If we **fail** to obtain additional capital, we may be required to significantly reduce, or refocus, our operations and our business, results of operations and financial condition could be materially and adversely effected. (ANDREA ELECTRONICS CORP, 2001/03/30)
failur*	A **failure** of the Company to improve its accounts receivable collections from the State of California could have a material adverse effect on the Company's business, financial condition and operating results. (CURATIVE HEALTH SERVICES INC, 2006/04/11)
impair*	Our investment in DISC common stock at October 31, 2009 is recorded at fair value of approximately $198,000 including a write-down of approximately $124,000 as a result of an other than temporary **impairment**, and has exposure to price risk. (COPYTELE INC, 2010/01/29)
inconsist*	Acquisition candidates brought to our attention through Mr. Penteliuk may include clients of Genuity Financial Group or its subsidiaries and Genuity Financial Group and its subsidiaries may have contractual, fiduciary or other obligations to these clients which may be **inconsistent** with, or adverse to, our interests and the interests of our stockholders. (TAILWIND FINANCIAL INC, 2008/09/15)
lose*	Qualifying a new contract manufacturer and commencing volume production can be expensive and time consuming. If we are required to change or choose to change contract manufacturers, we may **lose** revenue and damage our customer relationships. (MCDATA CORP, 2001/03/02)
loss*	Notwithstanding this, inflation can directly affect the value of loan collateral, in particular real estate. Sharp decreases in real estate prices have, in past years, resulted in significant loan **losses** and **losses** on real estate acquired. Inflation, or disinflation, could significantly affect NewMil's earnings in future periods. (NEWMIL BANCORP INC, 2002/03/29)

(*continued on next page*)

Table A.2 (*continued*).

Key Word Stem	Sentences
lost	We have experienced delays in the introduction of new products due to various factors, resulting in **lost** revenue. (NOVELL INC, 2007/05/25)
neg*	As a result of the activities of the FDA, the CDC and the Secretary's Advisory Committee on Genetic Testing, it is possible that our existing and future assays may be subject to a regulatory approval similar to the pre-marketing approval process which the FDA applies to drugs and medical devices, or may be subject to other increased regulatory standards, which could have a **negative** effect on our business. (SPECIALTY LABORATORIES, 2001/03/30)
nonperform*	Sotheby's counterparties to these sharing arrangements are typically international art dealers or prominent art collectors. Sotheby's could be exposed to losses in the event of **nonperformance** by these counterparties. (SOTHEBYS, 2009/02/27)
pressur*	These competitive **pressures** could result in increased pricing **pressures** on a number of our products and services, particularly as competitors seek to win market share, and may harm our ability to maintain or increase our profitability. (HARTFORD LIFE INSURANCE CO, 2007/02/23)
risk*	Because we market products to the public, we face an inherent business **risk** of financial exposure to product liability claims. As a publisher of online content, we face potential liability for defamation, negligence, copyright, patent or trademark infringement, or other claims based on the nature and content of materials that we publish or distribute. (SCIENTIFIC LEARNING CORP, 2002/03/29)
riski*	If we acquire additional businesses, these acquisitions will involve financial uncertainties as well as personnel contingencies, and may be **risky** and difficult to integrate. (PARLEX CORP, 2003/10/14)
slowdown*	The current economic **slowdown** has been more profound in the telecommunications market resulting in a significant reduction in capital expenditures for products such as our DWDMs and our fiber optic components. It is impossible to predict how long the **slowdown** will last.(APA OPTICS INC, 2003/07/01)
unabl*	If the Company loses the services of key personnel or if it is **unable** to attract additional qualified personnel, the Company's business and the price of its Common Stock could be materially and adversely affected. (CT COMMUNICATIONS INC, 2007/03/07)
weaken*	In fiscal 2001, **weakening** local currencies negatively impacted translated revenue compared to the prior year. Revenue in the region would have increased 6% if foreign exchange rates were constant with those of the prior year. (MICROSOFT CORP, 2001/09/18)
weaker	For example, higher gasoline prices in 2007 contributed to **weaker** demand in North America for certain vehicles for which we supply products, especially full-size SUVs and pick-up trucks. If gasoline prices remain high or continue to rise, the demand for such vehicles could weaken further and the recent shift in consumer interest to passenger cars and CUVs, in preference to SUVs and pick-up trucks, could be accelerated. (DANA HOLDING CORP, 2008/03/14)
weak*	As a result, our financial results could be significantly affected by factors such as changes in foreign currency exchange rates or **weak** economic conditions in the foreign markets in which we distribute our products. (INTERFACE INC, 2010/03/16)

This table presents examples of sentences in 10-K filings for downside risk related key word stems.

the current firm, the number of years the CEO has worked in the same firm and the number of working years in general. Finally, we incorporate CEO ownership as a large number of finance literature (see, for example, Jensen, 1986; Panousi and Papanikolaou, 2012; Nikolov and Whited, 2014) shows that managerial ownership is an important determinant of corporate investment, capital structure, cash holdings and payout policies. Our results indicate that the strong and persistent predictive power of the text measure remains unchanged after adding a large set of predictors capturing CEO characteristics.

Our final robustness check accounts for corporate hedging activities. Past work shows that some corporations manage their risk by implementing hedging to stabilize their earnings and cash flows, lower their bankruptcy risk and ease their credit and financial constraints. Managerial risk tolerance may also play a role in corporate hedging decisions (Bodnar et al., 2014). Depending on hedging activities, mangers facing risk changes may adjust corporate policies differently.

We construct a text-based measure of corporate hedging by counting the frequency of hedge-related word stems, including different forms of the word "hedge" and "derivative", in the 10-K reports. Our main variable, *Hedge*, is then computed as the frequency of hedge-related word stems divided by the total number of word stems in the 10-K reports. We find that an increase in hedging activities is associated with an increase in leverage and employment, and a decrease in cash holdings and dividend payouts. Ultimately, however, the predictive power of our text-based measure is robust to considering hedging. Both hedging and non-hedging firms significantly adjust their corporate policies subsequent to risk changes, yet hedging firms adjust their corporate policies more mildly.

7. Conclusion

This paper develops a methodology to measure firm-level downside risk through analyzing the textual contents of corporate 10-K reports. It then examines whether the text measure is useful in predicting changes in corporate policies including leverage, investment, employment, R&D, cash holdings, and payout.

Our new measure is based on keywords characterizing managerial perception of downside risk in 76,676 10-Ks for the period January 1994 through September 2015. Compared with stock return and earnings volatility, the text-based measure focuses exclusively on future downside possibilities and information related to firms' fundamentals. Furthermore, our dictionary is not limited to the word "risk" per se (Li, 2006), words with negative tones (Loughran and McDonald, 2011), or words characterizing product market competition and threat (Li et al., 2013; Hoberg et al., 2014). Rather, the measure aims to capture a comprehensive set of well-documented risk factors from macro conditions, financial market, counterparties, and day-to-day operations.

Our results based on OLS and logit regressions show that the new measure predicts consistent reductions in leverage, capital expenditure, R&D, employment, and dividend payouts and stock repurchases, along with increasing cash holdings. In contrast, the relation between changes in corporate policies and changes in stock return volatility and earnings volatility are mostly insignificant. We further use LASSO regressions to examine predictability of corporate policies based on a comprehensive set of predictors including existing text-based measures as well as various corporate control variables and macro variables. We find that the new text-based measure is among the most successful predictors compared with majority of corporate control variables such as firm size, profitability and Tobin's Q and macro variables such as S&P 500 index return and Treasury yield. The predictive power of our measure is even more striking compared with existing textual variables, stock return volatility and earnings volatility in terms of all corporate policies examined. Overall, our paper extends the growing textual analysis literature by showing that our text-based measure, which includes a broader set of downside risk-related words, has strong predictive power in corporate capital structure, investment, cash holdings and payout policies that goes beyond existing text-based and other variables.

Appendix A

See Tables A.1 and A.2.

Appendix B. Data construction

We construct a comprehensive set of alternative text-based proxies using dictionaries developed by previous studies. We develop textual variables based on the dictionary of the word "risk" per se (Li, 2006), the dictionary of risk-related keywords in Kravet and Muslu (2013), and the dictionary of positive, negative, uncertainty, litigation, strong modal, weak modal, and constraining words in Loughran–McDonald Sentiment Word Lists (Loughran and McDonald, 2011, 2013), respectively. We construct additional textual variables using the dictionary of positive or negative words from the Harvard Psychosociological Dictionary (Tetlock, 2007) and the combination of negative words from both the Loughran–McDonald Sentiment Word Lists and Harvard Psychosociological Dictionary. Finally, we adopt the text-based measure of product market fluidity in Hoberg et al. (2014).

We use book leverage ratio, measured by the ratio of total liabilities to total assets, as a proxy for capital structure. The main independent variable, active change in book leverage ($dlev_{t+1}$), is then computed as change in book leverage ratio from year t to $t+1$, excluding the effect of change in retained earnings. Following previous literature (see, for example, Harris and Raviv, 1991; Frank and Goyal, 2009), our control variables in capital structure regressions include: (1) lagged book leverage ratio; (2) change in stock return volatility; (3) logarithm of sales as a proxy for size; (4) annual stock return; (5) tangibility measured by the ratio of gross properties, plant, and equipment (*PPE*) to total assets; (6) market-to-book ratio as a proxy for growth; (7) EBIT divided by total assets (*ROA*) as a proxy for profitability; (8) effective corporate tax rate; (9) short-term solvency measured by the ratio of cash to interest expenses; (10) dividend yield measured by the ratio of common equity dividends to the market value of common equity; (11) external financing, measured by financial deficit normalized by sales.[27] We also control for macro conditions including annual S&P 500 value-weighted return, one-year constant maturity Treasury bill yield, default risk premium measured by the yield difference between Moody's Baa and Aaa indexes, *VIX*, and industrial production growth.

Next, we compute the percentage change in capital expenditure from year t to $t+1$, $\%dcapx_{t+1}$, as one primary policy variable in investment regressions. We also compute $\%demp_{t+1}$ as the percentage change in employment to proxy for investment in human capital. The percentage changes in R&D and advertising expenses, $\%dxrd_{t+1}$ *and* $\%dxad\ _{t+1}$, establish longer-term investment in R&D and product development. Following Panousi and Papanikolaou (2012), our control variables in predicting adjustments in capital expenditure, employment, and R&D policies are: (1) ratio of cash flow (measured as earnings before extraordinary items plus depreciation) to PPE; (2) Tobin's Q computed as the ratio of market value of assets to book value of assets; (3) change in stock return volatility; (4) market leverage, measured as total liabilities divided by the sum of total liabilities and market value of equity; (5) logarithm of sales; (6) *ROA*; (7) tangibility measured by the ratio of gross properties, plant, and equipment (*PPE*) to total assets; (8) market-to-book ratio as a proxy for growth.

We define change in cash holdings, $dcash_{t+1}$, as the change in the ratio of sum of cash, cash equivalent and short-term investments to total assets. Following previous literature (e.g., Bates et al., 2009; Hoberg et al., 2014; Gao et al., 2014), we incorporate the following control variables: (1) lagged cash as the sum of cash and short-term investments ($cash_{i,t}$); (2) lagged change in cash holdings ($dcash_{i,t}$); (3) ratio of net working capital to total assets; (4) *Dividend Dummy*, which equals one if common dividends are paid and zero otherwise; (5) ratio of R&D expenses to sales; (6) ratio of capital expenditure to total assets; (7) logarithm of sales; (8) change in stock return volatility; (9) book leverage ratio; (10) market-to-book ratio as a proxy for growth; (11) ratio of cash flow (measured as earnings before extraordinary items plus depreciation) to PPE.

As in Hoberg et al. (2014), we develop four measures to examine changes in dividend payout policy: (i) $Dividend\ Initiation_{i,t+1}$, which equals one if a firm initiates dividend payment and zero otherwise; (ii) $Dividend\ Omission_{i,t+1}$, which equals one if a firm

[27] We follow Chen et al. (2014) to compute financial deficit as the difference between cash outflow and internally generated cash flow. In particular, cash outflow includes investment in PPE and intangible assets, and increase in working capital. Internally generated cash flow is the sum of net income, depreciation and amortization and deferred tax minus dividend payouts.

Table C.1
Variable definition.

Variable	Definitions
Panel A: Text-based Variables	
Total Words	Total number of meaningful word stems in the 10k file.
Downside Risk Words	Total number of downside risk related word stems (shown in Appendix A) in the 10k file.
Risk Level	The ratio of downside risk-related words to total meaningful words in the 10K file.
Risk Change	Annual change in Risk Level.
Rising Risk	A dummy variable that equals one when the firm experiences rising risk in a magnitude larger than the median increase in risk level (i.e., 0.09%), and zero otherwise.
Resolving Risk	A dummy variable that equals one when the firm experiences decreasing risk in a magnitude larger than the median decrease in risk level (i.e., −0.06%), and zero otherwise.
dLi Risk	Annual change in the counts of the word "risk" per se (Li, 2006) divided by *Total Words.*
dKM Risk	Annual change in the counts of risk-related keywords in Kravet and Muslu (2013) divided by *Total Words.*
dLM Uncertainty	Annual change in the counts of uncertainty words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Negative	Annual change in the counts of negative words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Positive	Annual change in the counts of positive words in the Loughran–McDonald Sentiment Word Lists divided by Total Words.
dLM Litigation	Annual change in the counts of litigious words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Strong Modal	Annual change in the counts of strong modal words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Weak Modal	Annual change in the counts of weak modal words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Constraining	Annual change in the counts of constraining words in the Loughran–McDonald Sentiment Word Lists divided by *Total Words.*
dLM Hrvd Negative	Annual change in the counts of negative words in Loughran–McDonald Sentiment Word Lists or Harvard Psychosociological Dictionary divided by *Total Words.*
dHrvd Negative	Annual change in the counts of negative words in the Harvard Psychosociological Dictionary divided by *Total Words.*
dHrvd Positive	Annual change in the counts of positive words in the Harvard Psychosociological Dictionary divided by *Total Words.*
dFluidity	Annual change in product fluidity (Hoberg, Phillips, and Prabhala, 2014).
Panel B: Corporate Policy Variables	
Book Leverage	Total liabilities/total assets.
dlev	Total liabilities$_t$/(total assets$_t$-(retained earnings$_t$- retained earnings$_{t-1}$))- total liabilities$_{t-1}$/total assets$_{t-1}$
%dDebt	(Total liabilities$_t$-total liabilities$_{t-1}$)/total liabilities$_{t-1}$
%dEquity	(Shareholder's equity$_t$-shareholder's equity$_{t-1}$-(retained earnings$_t$- retained earnings$_{t-1}$)) /shareholder's equity$_{t-1}$
Capital Expenditure	Capital expenditure in million dollars.
%dcapx	Percentage change in capital expenditure.
Employment	Number of employees in thousand dollars.
%demp	Percentage change in Employment.
R&D	Research and development expenses. We replace missing values with zero.
%dxrd	Percentage change in R&D.
Advertising Expense	Research and development expenses. We replace missing values with zero.
%dxad	Percentage change in advertising expense.
Cash	Cash and short-term investments in million dollars.
Cash/Assets	Cash and short-term investments divided by total assets.
dcash	Annual change in Cash/Assets.
Dividends (per share)	Dividends declared on common equities per share.
Dividend Initiation	A dummy variable that equals one if the company initiates dividends and zero otherwise.
Dividend Omission	A dummy variable that equals one if the company omits dividends and zero otherwise.
Dividend Decrease	A dummy variable that equals one if the company decreases dividends and zero otherwise.
Dividend Increase	A dummy variable that equals one if the company increases dividends, and zero otherwise.
Net Repurchases	Purchases of common and preferred stock (Compustat item "prstkc") less the reduction in the value of preferred stocks outstanding (Compustat item "pstkrv").
Repurchase More than 1% Asset Dummy	An indicator variable that equals one if Net Repurchases is more than 1% of total assets and zero otherwise.
Panel C: Corporate Control Variables	
Assets	Total assets in million dollars.
Capital Expenditure/Assets	Capital expenditure divided by total assets.
Cash	Cash and short-term investments in million dollars.
Cash Flow/PPE	Earnings before extraordinary items plus depreciation normalized by the amount of property, plant, and equipment.
Cash/Interest Expenses	Cash and short-term investments divided by interest expense.
Credit Rating	An indicator variable for S&P Domestic Long Term Issuer Credit Rating. It ranges from 2 (for "AAA" rating) to 29 (for "SD" rating).
Dividend Dummy	An indicator variable that equals one if the company pays dividends and zero otherwise.
Dividend Yield	Dividends per share divided by fiscal year end stock market price.
Earnings Volatility	Standard deviation of operating income/total asset using past five-year data.

(*continued on next page*)

Table C.1 (*continued*).

Variable	Definitions
Effective Tax Rate	Income tax divided by pretax income.
Financial Deficit/Sales	Difference between cash outflow and internally generated cash flow. Cash outflow includes investment in PPE, intangible assets, and increase in net working capital. Internally generated cash flow includes net income plus depreciation and amortization and deferred tax minus dividends.
Firm Age	Number of years since date of IPO.
Market Leverage	(total assets - stockholders' equity)/ (total assets - stockholders' equity + market value of equity).
M/B Ratio	Market value of equity divided by book value of equity.
Neg. Earn. Dummy	A dummy variable that equals one if net income is negative, and zero otherwise.
Net Working Capital/Assets	Net working capital divided by total assets.
R&D/Sales	R&D divided by net sales.
Retained Earnings/Assets	Retained earnings divided by total assets.
ROA	EBIT divided by total assets.
Sales	Net sales in million dollars.
Sales Growth	Percentage of change in net sales.
Stock Return	Annualized daily stock return in the fiscal year.
Stock Return Volatility	Annualized daily stock return volatility including dividend.
Change in Volatility	Annual change in stock return volatility from the previous year.
Tangibility	Net property, plants and equipment's divided by total assets.
Tobin's Q	The sum of total liabilities and market value of equities divided by book value of assets.
Panel D: Macroeconomic Variables	
Default Spread	Default Spread between Baa and Aaa rated Bonds.
Industrial Production Growth	Industry production growth from one year ago.
Treasury Bill Yield	Constant 1-year Maturity Treasury Bill Rate.
S&P 500 Return	Annual return of S&P 500 index.
VIX	CBOT option-implied annualized volatility.

terminates dividend payment and zero otherwise; (iii) *Dividend Increase*$_{i,t+1}$, which equals one if a firm increases dividend payment and zero otherwise; (iv) *Dividend Decrease*$_{i,t+1}$, which equals one if a firm decreases dividend payment and zero otherwise. The first two measures capture abrupt changes in dividend policy, while the follow-ups reflect moderate adjustment in payouts. Following Hoberg et al. (2014) and Farre-Mensa et al. (2014), we consider the following control variables in the dividend policy regressions: (1) logarithm of firm age as the number of years since the date of IPO; (2) sales growth proxied by percentage change in net sales; (3) *Negative Earnings Dummy* equals one if net income is negative and zero otherwise; (4) ratio of retained earnings to total assets as a proxy for firm maturity; (5) logarithm of sales; (6) ratio of R&D to sales; (7) market-to-book ratio; (8) change in stock return volatility; and (9) ROA.

We construct an indicator variable, *Repurchase More than 1% Asset Dummy*, which equals one if the value of net stock repurchases is over 1% of total assets and zero otherwise. Following Hoberg et al. (2014), we define the value of net repurchases as purchase of common and preferred stocks less the reduction in the value of outstanding preferred stocks. We also use the value of net repurchases as an alternative policy variable. We use the same set of control variables as in the dividend policy regressions.

Appendix C

See Table C.1.

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